

Calculus And Linear Algebra

Sem 1

Watermarked study resource

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CATEGORY: PYQs

STATUS: ok

[Page 1]

Date: 19.09.2025 Total Marks: 50

Q. No

1

No

A=-2 2

5

SRM Institute of Science and Technology Department of Mathematics

2 (a) Find the eigenvalues of the matrix $\begin{pmatrix} -1 & 0 \\ 0 & -2 \end{pmatrix}$

Find the eigenvalues and eigenvectors of the matrix $\begin{pmatrix} 10 & -2 \\ -5 & 1 \end{pmatrix}$

21MAB101T-Calculus and Linear Algebra

Part-A (3*8-24 Marks) (Answer any THREE Questions)

Academic Year: 2025-2026 -ODD

dz

$\frac{dy}{dx}$

A by using the property.

dt

FT-II

3 Expand $x^2y + 3y^2$ in powers of $(x-1)$ and $(y+2)$ using Taylor's series up to second degree terms.

-3 0 and hence find the eigenvalues of $\begin{pmatrix} 4 & 2 \\ 3 & 0 \end{pmatrix}$

4 (a) If $z = xy + xy^2$, where $x = at$, $y = 2at$, then find without direct substitution.

(b) Discuss the nature of the Quadratic form $x^2 + 3x^2 + 6x + 2x^2 + 2x^3 + 2x^2 + 4x^3$, without reducing to the canonical form.

y

(b) If $u =$ and $v =$ then find)

Part-B (2*13-26 Marks) (Answer any TWO Questions) Reduce the Quadratic form $3x^2 + 5x^2 + 3x^2 + 2x^2 + 2x^2 + 2x^2 + 4x^3$, into canonical form by an orthogonal transformation and determine the signature, rank and nature of the Quadratic form.

T34S10

Time: 08.00-09.40 AM

Duration: 1 hour 40 Minutes

8

Marks BL CO PO

4

8

4

13

2

2

2

B2

3

Set-B

2

4

1

1

2

2

3 2

2

1,2

1,2

1,2

1,2

MarksBL CO PO

12

1,2

[Page 2]

6 Verify Cayley Hamilton theorem for the matrix 21

No

2(a)

Register No.:

2(b)

Name of the Student:

3

4(a)

4(b)

5

-(1 2

6

7

Find the volume of the largest rectangular parallelopiped that can be inscribed in the ellipsoid $x^2 + y^2 + z^2 = 1$.

C01 c01 C03 Co4 C0s C0s

C0

1
1
2
2
2
1 -1 2
1
1
2

Part-A (3*8=24 Marks) Allotted Awarded Marks

8
Course Outcome (CO) and Bloom's level (BL) Coverage in Questions
4
4
8
and hence find 4*.

4
4
Part-B (2-13=26 Marks) 13

13
Marks
Evaluation Sheet
Total

CO
Consolidated Marks:

CO1
CO2
13
Total

13
231
Blooms level

3
Marks Scored

4
Signature of the Course Teacher

1
2
1,2
1,2

[Page 3]
Date: 19.09.2025
Total Marks: 50
Course Outcomes (CO

COT Analy he con

Q. No

SRM Institute of Science and Technology Department of Mathematics Academic Year: 2025-2026

ODD

TAeey fh cac MeaENlg ao fspiay pdam

Q. No

21MAB101T-Calculus and Linear Algebra

MFT anf Eae Vectun rbes 4ging ScCE nErgeg

Lonhae, ant Ter sefeu ale pbas Soe and Egihog

1 Find the eigenvalues and eigenvectors of the matrix

$\begin{pmatrix} 1 & 3 \\ 3 & 5 \end{pmatrix}$

$A = \begin{pmatrix} -2 & -3 \\ 3 & 5 \end{pmatrix}$

FT-II

Nt cpe ign of So and Engherg

Part-A (3*8=24 Marks)

(Answer any THREE Questions)

2(a) If 2 and 3 are the eigenvalues of the matrix

$\begin{pmatrix} 3 & 10 \\ 5 & 3 \end{pmatrix}$

$\begin{pmatrix} 3 & 5 \\ 3 & 5 \end{pmatrix}$

find the third eigen value of A and

hence find the eigenvalues of the matrix A- by using property.

3 Expand $e^{\sin y}$ in powers of $(x-1)$ and $(y -;)$ using Taylor series up to second degree terms.

4 (a) If $x^2 + y^2 - 2xy = 1$, then find

(b) Discuss the nature of the Quadratic form $3x^2 + 3xy + 5x - 2y$

$-6x^2 + 6xy - 6y^2$ without reducing to

the canonical form.

$\frac{dy}{dx}$

$\frac{dx}{dy}$

(b) If $x = u^2 - v^2$ and $y = 2uv$, then find

Part-B (2*13=26 Marks)

0(:9)

(Answer any TWO Questions)

5 Reduce the Quadratic form $22x^2 + x^2 + x^2 + 2x, x, -$

$4x^2 + 3x, - 2x^2 + 3x$ into canonical form by an orthogonal

transformation and determine the signature, rank and nature of the Quadratic form.

Time: 08.00-09.40 AM Duration: 1 hour 40 Minutes

Marks BL COPO

4

8

4

4

8

13

2

3

B2

2

3

4

Set-A

2

2 1 1,2

1 1,2

1 1,2

2 1,2

2 1,2

Marks BL CO PO

2 1,2

1,2

[Page 4]

6 Verify Cayley Hamilton theorem for the matrix

30

No

2(a)

Register No.:

2(b)

3

4(a)

Name of the Student:

c0Y

4(b)

5

6

Find the dimensions of the rectangular box, open at the top of maximum capacity whose surface area is 432 sq. cm.

CO

1

1

c0z c03 co4 C0S

2

3

2

-1

2

1

2

8

Course Outcome (CO) and Bloom's level (BL) Coverage in Questions

4

Part-A (3*8=24 Marks) Allotted Awarded Marks Marks

4

3

4

4 and hence find A-,

13

13

Part-B (2*13=26 Marks)

13

cos

Evaluation Sheet

Total

CO C01

Consolidated Marks:

C02

13

Total

13

31 1,2

Blooms level

BLA, 239%

Marks Scored

Signature of the Course Teacher

2 12

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[usp=drive_link&oid=104856686103320873965&rtpof=true&sd=true](https://docs.google.com/document/d/1uNgufPfaYgLVMrmx0_ETAkYyoju7L57R/edit?usp=drive_link&oid=104856686103320873965&rtpof=true&sd=true)

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Multiple Choice Questions

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Multiple Choice Questions

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Unit - I: Matrices

PART A

MULTIPLE CHOICE QUESTIONS

1. The matrix of the quadratic form

$1x^2 + 5x^2 + 2x^2 + 2xy + 2xz + 6xz + 2yz + 2z^2$ is

$\begin{pmatrix} 1 & 1 & 3 \\ 1 & 2 & 3 \\ 1 & 2 & 3 \end{pmatrix}$

(a) $\begin{pmatrix} 1 & 5 & 1 \\ 1 & 5 & 2 \\ 4 & 5 & 3 \end{pmatrix}$ (b) $\begin{pmatrix} 4 & 5 & 3 \\ 4 & 5 & 4 \end{pmatrix}$

(c) $\begin{pmatrix} 3 & 1 & 1 \\ 3 & 2 & 1 \\ 4 & 3 & 1 \end{pmatrix}$ (d) $\begin{pmatrix} 3 & 4 & 1 \\ 3 & 4 & 1 \end{pmatrix}$

☐ (a) ☐ (b) ☐ (c) ☐ (d)

☐ (a) ☐ (b) ☐ (c) ☐ (d)

☐ (a) ☐ (b) ☐ (c) ☐ (d)

☐ (a) ☐ (b) ☐ (c) ☐ (d)

☐ (a) ☐ (b) ☐ (c) ☐ (d)

a b c d

2. The number of positive terms in the canonical form is called

(a) Signature (b) Index (c) Quadratic form (d) Positive de finite

3. A homogeneous polynomial of second degree in any number of variables is

(a) Canonical form (b) Quadratic form (c) Orthogonal (d) Diagonal form

4. Find the eigen values of $2A$ if

$A = \begin{pmatrix} 3 & 2 & 4 \\ 0 & 2 & 6 \\ 0 & 0 & 5 \end{pmatrix}$

$A = \begin{pmatrix} 0 & 2 & 6 \\ 0 & 0 & 5 \end{pmatrix}$

$\begin{pmatrix} 0 & 0 & 5 \end{pmatrix}$

$\begin{pmatrix} 0 & 0 \end{pmatrix}$

$\begin{pmatrix} 0 & 0 & 0 & 0 \end{pmatrix}$

$\begin{pmatrix} 0 & 0 \end{pmatrix}$

$\begin{pmatrix} 0 & 0 \end{pmatrix}$

(a) 6, 4, 10 (b) 9, 4, 25 (c) 9, 2, 5 (d) 3, 2, 5

5. Find the sum and product of the eigen values of

$A = \begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 0 & 0 & 1 \end{pmatrix}$

$A = \begin{pmatrix} 1 & 2 & 1 \\ 0 & 0 & 1 \end{pmatrix}$

$\begin{pmatrix} 0 & 0 & 1 \end{pmatrix}$

$\begin{pmatrix} 0 & 0 \end{pmatrix}$

$\begin{pmatrix} 0 & 0 & 0 & 0 \end{pmatrix}$

$\begin{pmatrix} 0 & 0 \end{pmatrix}$

$\begin{pmatrix} 0 & 0 \end{pmatrix}$

(a) 5, 3 (b) 3, 5 (c) 2, 1 (d) 0, 1

6. The eigen values of an orthogonal matrix have the absolute value _____

(a) 0 (b) 1 (c) 2 (d) 1

7. All the eigen values of a symmetric matrix with real elements are

(a) Distinct (b) Real (c) Equal (d) Conjugate complex numbers

8. Find the nature of the quadratic form $2x^2 + 3y^2 + 2z^2 + 2xy + 2yz + 2zx$

(a) Positive definite (b) Negative definite (c) Positive semi-definite (d) Indefinite

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[Page 3]

9. Write the Q.F. defined by the matrix

$\begin{bmatrix} 6 & 1 & 7 \\ 1 & 2 & 0 \\ 7 & 0 & 1 \end{bmatrix}$

$A = \begin{bmatrix} 6 & 1 & 7 \\ 1 & 2 & 0 \\ 7 & 0 & 1 \end{bmatrix}$

$\begin{bmatrix} 6 & 1 & 7 \\ 1 & 2 & 0 \\ 7 & 0 & 1 \end{bmatrix}$

$\begin{bmatrix} 6 & 1 & 7 \\ 1 & 2 & 0 \\ 7 & 0 & 1 \end{bmatrix}$

$\begin{bmatrix} 6 & 1 & 7 \\ 1 & 2 & 0 \\ 7 & 0 & 1 \end{bmatrix}$

$\begin{bmatrix} 6 & 1 & 7 \\ 1 & 2 & 0 \\ 7 & 0 & 1 \end{bmatrix}$

$2x^2 + 3y^2 + 2z^2 + 2xy + 2yz + 2zx$

$1x^2 + 2y^2 + 3z^2 + 1xy + 2yz + 3zx$

$2x^2 + 3y^2 + 2z^2 + 2xy + 2yz + 2zx$

$1x^2 + 2y^2 + 3z^2 + 1xy + 2yz + 3zx$

(a) $6x^2 + 2x^2 + 14x^2$ (b) $6x^2 + 6x^2 + 7x^2$

(c) $6x^2 + 2x^2 + 14x^2$ (d) $6x^2 + 6x^2 + 14x^2$

$\begin{bmatrix} 6 & 1 & 7 \\ 1 & 2 & 0 \\ 7 & 0 & 1 \end{bmatrix}$

$\begin{bmatrix} 6 & 1 & 7 \\ 1 & 2 & 0 \\ 7 & 0 & 1 \end{bmatrix}$

10. Find the eigen values of the matrix

$\begin{bmatrix} 12 & 21 \\ 21 & 12 \end{bmatrix}$

$\begin{bmatrix} 12 & 21 \\ 21 & 12 \end{bmatrix}$

$\begin{bmatrix} 12 & 21 \\ 21 & 12 \end{bmatrix}$

$\begin{bmatrix} 12 & 21 \\ 21 & 12 \end{bmatrix}$

(a) 1, 3 (b) 3, 1 (c) 2, 1 (d) 1, 2

11. Find the eigen values of $10A$ if $12A = 0$

$\begin{bmatrix} 10 & 12 \\ 12 & 10 \end{bmatrix}$

$\begin{bmatrix} 10 & 12 \\ 12 & 10 \end{bmatrix}$

10 (a) 1, 3 (b) 3, 1 (c) 3, 1 (d) 0, 2

12. If the sum of two eigen values and trace of a 3×3 matrix A are equal, find the value of A

(a) 0 (b) 1 (c) -1 (d) 2

□□ □□□

☐ ☐☐ ☐ ☐☐ ☐ ☐☐ ☐ ☐☐

cd

1 2 3 1 2 2 3 3 1x 2x 3x 2x x 2x x 2x x ? ? ? ? ?

□ □ □ □ □ □

16. The matrix of the quadratic form $2x^2 + 4xy + 2y^2$ is

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[illegible]

a b c d

201

A 0 2 0

102

??

?????

??

??

are 1 and 2. Find the third eigen value.

(a) 3 (b) b (c) 2 (d) 1

18. Two of the eigen values of 3 x 3 matrix A are 2, 1 and $|A| = 12$. Find the third eigen value

(a) 6 (b) 3 (c) 2 (d) 1

19. If A is an orthogonal matrix then

(a) $|A| = 0$ (b) A is singular (c)

$2AI = I$ (d) $T1AA = I$

20. Two eigen values of

4 6 6

A 1 3 2

1 5 2

??

?????

??? ? ???

are equal and they are double the third. Find them.

(a) 1, 2, 2 (b) 2, 1, 1 (c) 2, 0, 1 (d) 1, 2, 3

21. Find the inverse of the eigen values of the matrix if $12A = 54$

??????????

() 1, 1/6 () 1, 1/6 () 1, 1/6 () 1, 1/6 ? ? ? ? a b c d

22. Find rank and index of the QF whose canonical form is

22

23

3y 3y?

(a) 2, 1 (b) 1, 2 (c) 0, 1 (d) 0, 2

23. Find signature of the QF whose canonical form is

222

1 2 32y y y?? ,

(a) 1 (b) -1 (c) 0 (d) 6

24. The eigen vectors corresponding to the distinct eigen values of a real symmetric matrix are

(a) imaginary (b) non-orthogonal (c) real (d) orthogonal

25. Find the characteristic equation of the matrix $01A = 23$

??????????

2 2 2 2(a) 3 2 0 (b) 3 2 0 (c) 3 3 0 (d) 6 3 0 ? ? ? ? ? ? ? ? ? ? ? ? ? ? ? ?

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[Page 5]

UNIT-II-FUNCTIONS OF SEVERAL VARIABLES

1. If $Z =$

$+3$

then what is

?

(i) $2y+3x$ (ii) $3y$ (iii) $2x+3y$ (iv) $2x$

2. $u =$

is homogeneous function of degree

(i) 2 (ii) 3 (iii) 1 (iv) 4

3. If $u = a$

$+2hxy+b$

then using Euler's theorem find x

$+ y$

$= ?$

(i) u (ii) $2u$ (iii) $3u$ (iv) $n(n-1)$

4. If $f(x, y) =$

then what is

$(1, 1) ?$

(i) $-e$ (ii)

(iii) e (iv) $-$

5. if $z = \log ($

$+xy+$

$)$ then what is x

$+y$

$=?$

(i) 1 (ii)

(iii) 2 (iv)

6. If $f(x, y)$ is an implicit function then

$=?$

(i)

(ii)

(iii)

(iv)

7. If $f(x, y) =$

?

(i) 1 (ii) -1 (iii) 0 (iv) 2

8. If $f(x, y) = \cos x \cos y$ then

$(0, 0) = ?$

(i) 1 (ii) 0 (iii) -1 (iv)

9. If $f(x, y) =$

then

$(1, 1)$ is

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[Page 6]

(i)

(ii) $\frac{1}{2}$ (iii) $-\frac{1}{2}$ (iv) 0

10. If $rt =$

at (a, b) then the point is

(i) Maximum point (ii) minimum point (iii) saddle point (iv) none of the se

11. The stationary points of

are

(i) $(-3, 0)$ (ii) $(0, 3)$ (iii) $(0, -3)$ (iv) $(3, 0)$

12. If $x =$

and $y = 2uv$ then J

is

(i)

(ii) 2

(iii) 4

(iv) 4

13. If $x = r$

and $y = r$

Then what is

$= ?$

(i)

(ii) r (iii) $2r$ (iv) 0

14. If $v =$

$+$

then

is

(i) $1+$

(ii)

(iii)

(iv) $1+$

15. u and v are functionally dependent if their jacobian value is

(i) zero (ii) one (iii) non-zero (iv) greater than zero

16. if

and

$=$

then

=?

(i) 0 (ii) 1 (iii) -1 (iv) 2

17. The stationary points of $f(x, y) =$
are

(i)

(ii)

18. The point $(0, 0)$ for $f(x, y) =$

is

(i) a maximum point (ii) a minimum point (iii) a saddle point (iv) none of these

19. If $f(x, y) =$

where $x = r \cos$

and $y = r \sin$

then

is

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(i) r (ii)

(iii) 1 (iv) 0

20. If $f(x, y) =$

$y + \sin y +$

then

(1,

) is

(i) 2

$-e$ (ii) 2π (iii) $2\pi + e$ (iv) 0

21. $u =$

is homogeneous function of degree

(ii) 1 (iii) 2 (iv) 3

22. If $u =$

then x

$+ y$

$= ?$

(i) $\sin u$ (ii) $\cos u$ (iii) $\sin 2u$ (iv) $\tan u$

23. If $u =$

, $v =$

then

$= ?$

(i) -3 (ii) 3 (iii) -

(iv)

, $y = r$
 $z = z$ then
 $= ?$
 (i) $2r$ (ii)
 (iii)
 (iv) r
 25. If $u =$
 and $v = x + y$ then
 $= ?$
 (i) $2x$ (ii) $2x + 2$ (iii) $2y - 2$ (iv) $2x - y$

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[Page 8]

ANSWERS

1. (iii) $2x + 3y$
2. (iii) 1
3. (ii) $2u$
4. (iii) e
5. (iii) 2
6. i)
7. (iii) 0
8. (iii) - 1
9. (ii) $\frac{1}{2}$
10. (iii) saddle point
11. (i) $(-3, 0)$
12. (iii) 4(
13. (ii) r
14. (iii)
15. (i) zero
16. (ii) 1
17. (ii)
18. (iii) a saddle point

19. (iv) θ

20. (iii) $2\pi + e$

21.

22. (iii) $\sin 2u$

23. (iii) -

24. (iv) r

25. (ii) $2x+2$

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MA1001- CALCULUS AND SOLID GEOMETRY

Unit-III Ordinary Differential Equations

Multiple Choice Questions

1. Which of the following is the general solution to $\frac{d^2y}{dx^2} + 3 \frac{dy}{dx} - 10y = 0$

(a) $y = Ae^{2x} + Be^{-5x}$

(b) $y = Ae^{-2x} + Be^{5x}$

(c) $y = Ae^{-2x} + Be^{-5x}$

(d) $y = Ae^{2x} + Be^{5x}$

2. Solution of $(D^2 + 4)y = 0$ is

(a) $y = A \cos 2x + B \sin 2x$

(b) $y = Ae^{2x} + Be^{-2x}$

(c) $y = A \cos$

$\sqrt{2}x + B \sin$

$\sqrt{2}x$

(d) $y = (Ax + B)e^{2x}$

3. The P.I of $(D^2 + 4)y = \sin 2x$ is

(a) $-x$

(b) $4 \cos 2x$

(c) x

(d) $-x$

$2 \cos 2x$

4. The equation $(a_0x^2D^2 + a_1xD + a_2)y = Q(x)$ is called, where $a_0, a_1, a_2 \in \mathbb{C}$

(a) Cauchy's equation

(b) Legendre's equation

(c) Taylor's equation

(d) Clairaut's equation

5. Use the transformation $z = \log x$, convert the D.E $x^2y'' - xy' + y = x^2$ to an equation with constant coefficients

(a) $(\theta^2 - 2\theta + 1)y = e^{2z}$

(b) $(\theta^2 - 2\theta + 1)y = ez$

(c) $(\theta^2 + 2\theta + 1)y = e^{2z}$

(d) $(D^2 + 2D + 1)y = ez$

6. The solution of $(D^2 + 2D + 1)y = 7$ is

(a) $y = (Ax + B)e^{-x} + 7$ (b) $y = (Ax + B)e^{-x} - 7$ (c) $y = (Ax + B)e^x + 7$

(d) $y = (Ax + B)e^x - 7$

7. The P.I of $(D - 1)^2y = ex \sin x$ is

(a) $-ex \cos x$ (b) $ex \cos x$ (c) $ex \sin x$ (d) $-ex \sin x$

8. The P.I of $(D - 1)^2y = x$ is

(a) $2 - x$ (b) $x + 2$ (c) x^2 (d) $-x^2$

9. If $1 \pm 2i$ are the roots of A.E of a differential equation $f(D)y = 0$ then the general solution

is

(a) $e^{-2x} (A \cos x - B \sin x)$ (b) $Ae^x + Be^{-2x}$ (c) $ex (A \cos 2x + B \sin 2x)$ (d) $Ae^t + Be^{2x}$

10. Convert the equation $(5 + 2x)^2y$

''

$- 6(5 + 2x)y$

,

$+ 8y = 0$ to an equation with constant

coefficient by using the transformation $z = \log(5 + 2x)$

(a) $(D^2 + 4D + 2)y = 0$ (b) $(D^2 - 4D + 2)y = 0$ (c) $(D^2 + 4D + 4)y = 0$ (d) $(D^2 + 4D - 2)y = 0$

11. The P. I of $(D^2 + 4)y = \sinh 2x$ is

(a) $yp = \sinh 2x$

8 (b) $yp = \sinh 2x$

4 (c) $yp = -\sinh 2x$

8 (d) $yp = -\sinh 2x$

4

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2

12. The P.I of $(D^2 + 6D + 5)y = e^{-x}$ is

(a) $yp = xe^{-x}$

4 (b) $yp = xe^{-x}$

2 (c) $yp = e^{-x}$

2 (d) $yp = e^{-x}$

4

13. The solution of $(D^2 - 2aD + a^2)y = 0$ is

(a) $Ae^{ax} + Be^{bx}$ (b) $Ae^{ax} + Be^{-ax}$ (c) $(Ax + B)e^{ax}$ (d) $(Ax + B)e^{-ax}$

14. The P.I of $(D^2 + 16)y = \cos 4x$ is

(a) x

2 $\sin 2x$ (b) $x \sin 4x$

8 (c) x

2 $\cos 2x$ (d) $x \cos 4x$

8

15. The C.F of $D^2y + y = \operatorname{cosec} x$ is

(a) $Ae^{ax} + Be^{bx}$ (b) $A \cos x + B \sin x$ (c) $(Ax + B)e^{ax}$ (d) $(Ax + B)e^{-ax}$

16. If $y_1 = \cos ax$, $y_2 = \sin ax$ then the value of $y_1 y_2$

,

$2 - y_2 y_1$

,

1 is

(a) $-a$ (b) 0 (c) 1 (d) a

17. Solve $(D^2 + 1)y = 0$ given $y(0) = 0$, $y'(0) = 1$

(a) $y = \sin x$ (b) $y = \cos x$ (c) $y = A \cos x + B \sin x$ (d) $y = 0$

18. The P.I of $(D - 2)^2 y = e^{2x}$ is

(a) x^2

$2e^{2x}$ (b) x

$4e^{2x}$ (c) x^2

$2e^{-2x}$ (d) x^2

$2e^{-2x}$

19. The P.I of $(D^2 + 4)y = \sin(2x + 5)$ is

(a) $-x$

$2 \sin(2x + 5)$ (b) x

$4 \sin(2x + 5)$ (c) $-x$

$4 \cos(2x + 5)$ (d) x

$2 \cos(2x + 5)$

20. Solve $(x^2 D^2 + xD + 1)y = 0$ is

(a) $Ae^{az} + Be^{bz}$ (b) $A \cos z + B \sin z$ (c) $(Az + B)e^{az}$ (d) $(Az + B)e^{-az}$

21. The roots of the auxiliary equation $(m^2 - 4) = 0$ are

(a) ± 2 (b) $\pm 2i$ (c) $\pm$

$\sqrt{2}$

2 (d) $1 \pm 2i$

22. The solution of $(x^2 D^2 - 7xD + 12)y = 0$ is

(a) $Ae^{-2z} + Be^{6z}$ (b) $Ae^{2z} + Be^{-6z}$ (c) $Ae^{2z} + Be^{6z}$ (d) $Ae^{-2z} + Be^{-6z}$

23. If $y_1 = \cos x$, $y_2 = \sin x$ then the value of $y_1 y_2$

,

$2 - y_2 y_1$

,

1 is

(a) -1 (b) 0 (c) 1 (d) 1

2

24. If three roots of the auxiliary equation become equal to the real number a , then the corresponding C.F is

(a) $(Ax^2 + Bx + C)e^{ax}$ (b) $Ae^{ax} + Be^{ax} + Ce^{ax}$ (c) $Ae^{ax} + (B \cos ax + C \sin ax)$ (d) a

25. The values of e^{ax}

$D-a$

(a) xe^{ax} (b) e^{ax} (c) $x^2 e^{ax}$ (d) x^2

$2e^{ax}$

Answers:

1. a 2. a 3. a 4. a 5. a 6. a 7. d 8. b 9. c 10. b 11. a
12. a 13. c 14. b 15. b 16. d 17. a 18. a 19. c 20. b 21. b
22. c 23. c 24. a 25. a

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SRM UNIVERSITY

MA1001- CALCULUS AND SOLID GEOMETRY

Unit-IV Geometrical Applications of Differential Calculus

Multiple Choice Questions

1. If the radius of curvature and curvature of a curve at any point are ρ and κ respectively, then

(a) $\rho = -1$

κ (b) $\rho = \kappa$ (c) $\rho = -\kappa$ (d) $\rho = 1$

κ

2. The locus of center of curvature is called

(a) Involute (b) Evolute (c) Radius of curvature (d) Envelope

3. The envelope of the family of curves $A\alpha^2 + B\alpha + C = 0$ (α is parameter) is

(a) $B^2 + 4AC = 0$ (b) $B^2 - AC = 0$ (c) $B^2 + AC = 0$ (d) $B^2 - 4AC = 0$

4. The curvature of the straight line is

(a) 1 (b) 2 (c) -1 (d) 0

5. The radius of curvature of the curve $y = 4 \sin x$ at $x = \pi$

2 is

(a) 1

2 (b) -1

2 (c) 1

4 (d) -1

4

6. The envelope of $ty - x = at^2$, t is the parameter is

(a) $x^2 = 4ay$ (b) $y^2 = 4ax$ (c) $x^2 + y^2 = 1$ (d) $x^2 - y^2 = 1$

7. The curvature at any point of the circle is equal to - - - of its radius

(a) Square (b) Same (c) Reciprocal (d) constant

8. What is the radius of curvature at (4, 3) on the curve $x^2 + y^2 = 25$

(a) 5 (b) -5 (c) 25 (d) -25

9. What is the curvature of a circle of radius 3

(a) 3 (b) -3 (c) 1

3 (d) -1

3

10. Find the envelope of the curve $y = mx + a$

m where m is a parameter

(a) $y^2 - 4ax = 0$ (b) $y^2 + 4ax = 0$ (c) $x^2 + y^2 = 1$ (d) $xy = c^2$

11. The radius of curvature of $y = e^x$ at $x = 0$ is

(a) 2

✓

2 (b) $2\sqrt{2}$

2 (c)

✓

2 (d) $\sqrt{2}$

2

12. The radius of curvature of the curve $y = \log \sec x$ at any point of it is

(a) $\sec x$ (b) $\tan x$ (c) $\cot x$ (d) $\operatorname{cosec} x$

13. In an ellipse the radius of curvature at the end of which axis is equal to the semi latus rectum of

the ellipse

(a) Minor (b) Major (c) Vertical (d) Horizontal

14. The radius of curvature of the curve $x = t^2, y = t$ at $t = 1$ is

(a) 5

✓

5

2 (b)

✓

5

2 (c) 5

2 (d)

✓

5

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2

15. Evolute of a curve is the envelope of -- of that curve

(a) Tangent (b) Normal (c) Parallel (d) Locus

16. The evolute of the cycloid $x = a(\theta - \sin \theta), y = a(1 - \cos \theta)$ is

(a) Astroid (b) Parabola (c) Cycloid (d) Circle

17. A curve which touches each member of a family of the curves is called -- of that family

(a) Evolute (b) Envelope (c) Circle of curvature (d) Radius of curvature

18. The envelope of family of lines $y = mx + am^2$ (where m is the parameter) is

(a) $x^2 + 2ay = 0$ (b) $x^2 + 4ay = 0$ (c) $y^2 + 2ax = 0$ (d) $y^2 + 4ax = 0$

19. The envelope of the family of lines

$tx + yt = 2c$, t being the parameter is

(a) $x^2 + y^2 = c^2$ (b) $xy = c^2$ (c) $x^2y^2 = c^2$ (d) $x^2 - y^2 = c^2$

20. The radius of curvature at any point on the curve $r = e^\theta$ is

(a)

$$\sqrt{r^2 + \left(\frac{dr}{dy}\right)^2}$$

(b) $\frac{r}{\sqrt{r^2 + \left(\frac{dr}{dy}\right)^2}}$

(c) $\frac{r}{\sqrt{r^2 + \left(\frac{dr}{dy}\right)^2}}$

(d) $\frac{r}{\sqrt{r^2 + \left(\frac{dr}{dy}\right)^2}}$

21. The radius of curvature in Cartesian coordinates is

(a) $\rho =$

$$\frac{1}{\sqrt{1 + y'^2}}$$

$$\frac{1}{y'^2}$$

(b) $\rho =$

$$\frac{1}{1 - y'^2}$$

$$\frac{1}{y'^2}$$

(c) $\rho =$

$$\frac{1}{1 + y'^2}$$

$$\frac{1}{y'^2}$$

(d) $\rho =$

$$\frac{1}{1 + y'^2}$$

$$\frac{2}{y'^2}$$

$$y'$$

22. The radius of curvature in polar coordinates is

(a) $\rho =$

$$\frac{r^2 + \left(r \frac{dr}{dr}\right)^2}{r^2}$$

$$\frac{r^2 + \left(r \frac{dr}{dr}\right)^2}{r^2}$$

$$\frac{r^2 - r \frac{dr}{dr}}{r^2}$$

$$\frac{r^2 + 2\left(r \frac{dr}{dr}\right)^2}{r^2}$$

(b) $\rho =$

$$\frac{r^2 - \left(r \frac{dr}{dr}\right)^2}{r^2}$$

$$\begin{aligned} & , \\ &)^2 \\ &)^3 / 2 \\ & r^2 - rr \\ & , \\ & + 2(r \\ & , \\ &)^2 (c) \rho = \\ & (\\ & r^2 - (r \\ & , , \end{aligned}$$

$$\begin{aligned} &)^2 \\ &)^3 / 2 \\ & r^2 - rr \\ & , \end{aligned}$$

$$\begin{aligned} & + 2(r \\ & , \\ &)^2 \end{aligned}$$

$$\begin{aligned} & (d) \rho = \\ & (\\ & r^2 + (r \\ & , \end{aligned}$$

$$\begin{aligned} &)^2 \\ &)^3 / 2 \\ & r^2 - rr \\ & , , \end{aligned}$$

$$\begin{aligned} & + 2(r \\ & , \end{aligned}$$

$$)^2$$

23. The radius of curvature in parametric coordinates is

$$(a) \rho =$$

$$\begin{aligned} & (\\ & (x \\ & , \end{aligned}$$

$$\begin{aligned} &)^2 + (y \\ & , \end{aligned}$$

$$)^2$$

$$)^3 / 2$$

$$\begin{aligned} & x \\ & , \end{aligned}$$

$$\begin{aligned} & y \\ & , , \end{aligned}$$

$$\begin{aligned} & - y \\ & , \end{aligned}$$

$$x$$

'' (b) $\rho =$

(
(x
,
)² + (y
,
)²
)^{3/2}
x
,
y
,,

+ y
,

x
'' (c) $\rho =$

(
(x
,
)² - (y
,
)²
)^{3/2}
x
,
y
,,

- y
,

x
,,

(d) $\rho =$

(
(x
,
)² - (y
,
)²
)^{3/2}
x
,
y
,,

+ y
,

x
,,

24. The equation of circle of curvature at any point (x, y) with center of curvature x, y and with

radius of curvature ρ is

- (a) $(x + x)^2 + (y + y)^2 = \rho^2$ (b) $(x - x)^2 + (y - y)^2 = \rho^2$ (c) $(x - x)^2 - (y + y)^2 = \rho^2$
(d) $(x + x)^2 + (y + y)^2 = \rho$

Answers:

1. d 2. b 3. d 4. d 5. c 6. b 7. c 8. a 9. c 10. b 11. a 12. a
13. b 14. a 15. b 16. c 17. b 18. b 19. b 20. d 21. a 22. d 23.
a 24. b

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SRM INSTITUTE OF SCIENCE & TECHNOLOGY

FACULTY OF ENGINEERING AND TECHNOLOGY

DEPARTMENT OF MATHEMATICS

Unit -IV Geometrical Applications of Differential Calculus

(Beta ,Gamma Functions)

Multiple Choice Questions

1. The value of $\int_0^4 x^{\beta} dx$, β is -----

(a).

!7

36 (b).

!7

! 6 (c).

!8

! 4 ! 4 (d).

!7

! 3

2. The value of $\int_0^1 x^{\beta} dx$

$\frac{1}{\beta+1}$

$\frac{1}{\beta}$

$\frac{1}{\beta-1}$

$\frac{1}{\beta+2}$

2

3,2

5

β is -----

(a).

8

$\pi(b)$.

8

$\pi(c)$.

16

$\pi(d)$.

16

2π

3. $\mathbb{R}(\mathbb{R}, \mathbb{R})$ is equal to -----

(a).

)! (

! !

$n\ m$

$n\ m$

+

(b).

)! (

! !

$n\ m$

$n\ m$

-

(c).

) ($n\ m$

$n\ m$

+ Γ

$\Gamma\Gamma$ (d).

) ($n\ m$

$n\ m$

- Γ

$\Gamma\Gamma$

4. The value of $\mathbb{R}$

$\mathbb{R}$

$\mathbb{R}\mathbb{R}$

$\mathbb{R}$

$\mathbb{R}\Gamma\ 2$

1

is -----

(a). $\pi(b)$. $2\pi(c)$. $\pi(d)$. π^2

5.) 1 ($n\ n-\Gamma\Gamma$ is equal to -----

(a). $\mathbb{R}$

∞ -

+0

1

1 dx

$x\ n$

$(b).) 1 , () 1 (n n-\Gamma \beta (c).) 1 , 1 () 1 (n n--\Gamma \beta (d).) , 1 () 1 (n n-\Gamma \beta$
 6. $\frac{1}{2}$
 2
 0
 $\cos \sin$
 π
 $\theta \theta \theta dq p$ is equal to -----
 (a) . $\frac{1}{2}$
 $\frac{1}{2}$
 $\frac{1}{2}\frac{1}{2}$
 $\frac{1}{2}$
 $\frac{1}{2} + +$
 2
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 $1 q p$
 $\beta (b). \frac{1}{2}$
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7. $\frac{1}{2}$

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$3^4 \log(x)$ is equal to -----

(a).

525

6 (b).

625

6 (c).

!5

! 6 (d).

!6

! 5

8. The value of θ

π

$d\theta$

2

0

cot is -----

(a).

2

π (b).

2

π (c).

2

π (d).

2

π

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UNIT V

Sequence and Series

1. The series is convergent if

(a) $p=1$ (b) $p=0$ (c) $p>1$ (d) $p<1$

2. The series is divergent if

(a) $p>1$ (b) $p=0$ (c) $p\leq 1$ (d) $p<1$

3. If $\sum a_n$ is a series of positive term such that $\lim_{n \rightarrow \infty} a_n > 1$, then the series is

(a) Convergent (b) Divergent (c) Oscillating (d) Monotonic

4. If $\sum a_n$ is a series of positive term such that $\lim_{n \rightarrow \infty} a_n = l$, then the series is convergent if

(a) $l<1$ (b) $l=0$ (c) $l>1$ (d) $l=1$

5. The series is

(a) Convergent (b) Divergent (c) Oscillating (d) Monotonic

6. The series is

(a) Convergent (b) Divergent (c) Oscillating (d) Monotonic

7. By D'Alembert's Ratio test l , the series is convergent when

(a) $l < 1$ (b) $l = 0$ (c) $l > 1$ (d) $l = 1$

8. By Raabe's test l , the series is divergent when

(a) $l < 1$ (b) $l = 0$ (c) $l > 1$ (d) $l = 1$

9. The series is

(a) Convergent (b) Divergent (c) Oscillating (d) Monotonic

10. A series is said absolutely convergent if the series

(a) is convergent (c) is divergent

(c) is convergent (b) is divergent

11. A series is said conditionally convergent if the series

$\sum_{n=1}^{\infty} \frac{1}{n^p}$

$\sum_{n=1}^{\infty} \frac{1}{n^p}$

$\sum_{n=1}^{\infty} \frac{1}{n^p}$

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- (a) is convergent (b) is divergent (c) is convergent (d) is divergent
12. The series is
(a) Convergent (b) Divergent (c) Conditionally convergent (d) absolutely convergent
13. The series is
(a) Convergent (b) Divergent (c) Conditionally convergent (d) absolutely convergent
14. The series is
(a) Conditionally convergent (b) absolutely convergent (c) Convergent (d) Divergent
15. An absolutely convergent series is
(a) Conditionally convergent (b) absolutely convergent (c) Convergent (d) Divergent
16. The series is
(a) Conditionally convergent (b) absolutely convergent (c) Convergent (d) Divergent
17. The series is
(a) Convergent (b) Conditionally convergent (c) absolutely convergent (d) Divergent

ANSWERS

1. d 11. c

2. c 12. d

3. b 13. c

4. a 14. d

5. a 15. c

6. b 16. c

7. a 17. a

8. a

9. b

10. a

11. nu

12. nu

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16. nu

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STATUS: ok

[Page 1]

Reg. No B.Tech./M.Tech (Integrated) DEGREE EXAMINATION, JANUARY 2023 First Semester

21MAB101T-CALCULUS AND LINEAR ALGEBRA (For the candidates admitted from the academic year

2022-2023) Note: Part A should be answered in OMR sheet within first 40 minutes and OMR

sheet should be handed over to hall invigilator at the end of 44 minute. Part-B and Part -C

should be answered in answer booklet. () Time: 3 Hours Max. Marks: 75

Marka M Co o PART A (20 x 1 20Marks) Answer ALL Questions 1. The number of positive terms in

the canonical form is called (A) Signature (C) Quadratic form B) Index (D) Positive definite

(1 2 4 Find the eigen values of A if $A = \begin{pmatrix} 0 & 2 & 6 \\ 0 & 5 & 2 \end{pmatrix}$.

(A) 1,4,25 (C) 4,4, 25 (B) 2,4, 20 (D) 1,2,25 I 2 3. If A is an orthogonal matrix then A^j is

(A) (C) 1 (B) tI D) -1 4. Find the sum and product of the eigen values of the matrix $\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$

(1 2-1) $A = \begin{pmatrix} -20 & 45 & 0 \\ 0 & -1 & -10 \end{pmatrix}$ (A) -1, -10 (C) -1, 10 B) 1, -10 (D) 1, 10 I2 2 2 If $z = 23y^2 + Sy$, then find

(A) $4x+5y$ C) $4x+6y$ (B) $4x-5y$ D) $4x+6y + 5xy$ 2 2 6. $\frac{dy}{dx}$ If $f(x, y)$ is an implicit function,

then is $\frac{dx}{dy}$ (A) 1) f_{xy} (C) i (B) of_{xy} (D) $(/ \ 093F1-21MAB101T$ Page 1 of 4

[Page 2]

12 2 2 7. If $f(x, y) = r'y + \sin y + e$, then find $S, 1r)$ (A) 2 (C) $27+e$ B) $27-e$ (D) 0 121 8. If u

and v are functionally dependent, then their Jacobian value is (A) Zero (C) One (B) Positive

(D) Negative 12 3 1 9. Solution of $(D^* + 4)y = 0$ (B) $y = A \cos 2x + B \sin \sqrt{2}x$ (A) $y = (Ar + B)e$ (C)

$y = A \cos 2x + B \sin 2x$ D) $y = Ae^{x^2} + Be^{1131}$ 10. The value of $D-a$ (A) xe (B) a (C) (D)

e

I2 3 2 11. Solve $(D + 5D + 4)y = 0$ (A) $y = Ae^{x^2} + Be^{4x}$ (C) $y = Ae + Bex$ (B) $y = Ae + Be$ () $y = Ae + Be$

12 3 2 12. Find the Particular Integral of $(D-4)y = \cos 2x$

(A) $\cos 2x$ (8) $\sin 2x$ $\sin 8$ (C) $\cos 2x \cos 2x$ 8 D) $-\cos 2x$ 8 1 I4 1 13. The curvature of the

straight line is (A) 1 (C) 0 (B) 2 (D) -1 41 14. The envelope of the family of curves of the

form $Aa + Ba + C = 0$, where a is the parameter is (A) $B+AC = 0$ (C) $B+4AC = 0$ (B) $B-4AC = 0$ (D) B

$-AC = 0$ 114 1 15. The radius of curvature in polar co-ordinates is 2 (A) $p = r + rP^* / -m + 2r$ (B) $p -$

- "I2- *+2 C)p=-(r7T2/2-r*" +2 (D) p=+r"2/2+r*+2

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[Page 3]

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[Page 4]

21 3 that .Ifu =u(x,y) and r=e cos6 and y=esin0 b. If show

ue r 3 3 1 8 23. a. Solve (D2+2D+1)y=* + sin 2x. (OR) 32 D.Solve (D +1)y= tan.x by the method of variation of parameters. 3 4 2

Find the radius of curvature of the curve r = all + cos6) at the point 0-(OR) 4 2 4 . Find the evolute of the parabola y = 4ax. 8 452 8 25. a. Test the convergence of the series Vn+1-Vn*-1.

2 2.4 2.4.62.4.6.8 0. Test the convergence of the Serncs 43.56 3.5.7.8 3.5.7.9.10 (OR) 4 5 2 8 +.00.

Marks BL PO PART-C(1 x 15 = 15 Marks) Answer ANY ONE Question 15 4 I 26. Reduce the quadratic form 3x-2y-z-4xy +8xz +12yz to canonical form by an orthogonal transformation. Discuss the nature of the quadratic form and also find rank, index, and signature.

2 2 27. Find the dimensions of the rectangular box open at the top, of maximum 15 42 2 capacity whose surface area is 432 sq.cm.

*** *

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CATEGORY: PYQs

STATUS: ok

[Page 1]

27. Testttre convergenoe ofthe series i,,, 1

"11@+1)Qn+1)

PART - C (5 x t2= 60 Marks)

AnswerAII euestions

28' a' vsrig, Cavlev-Hamlton theorem for rhe marix

^

=1j ; ll *a hence find A-r.tt(1 2 t)

b' Reduce rhe *:gr:, form 6fi fi;i:'G -4\pz-Zxzxs+44x1to canonicar form byan orthogonal transforrrration. Also state its nature.-

29' a Explain e'cosy in powers ofx andy as far as the terms of the third degree.

b' Find the volume of the larges, ,""[lfi* parallelepiped that can be inserted in the*2 1" ellipsoid *)* ,,!a.cr- b" c,

30' a' 5o1r. 1o2 -lo+12)y=e5, +cos2x.

b. solu" (*2D2+xD-g)y-+. - (oR)

x'

3 1 ' a' p;n6 the equation of the circle of curvature of the parabo la y2 - 12*at the point

(3, 6).

b' rirra the evolute of the parabola, $z = (or)$

32. a-i. State the axioms of convergence of series by comparison test.

ii. Test the convergence of the series $1 + 3.4 + 3.n.5 + \dots$

4 4.6 4.6.8 --- '

b.i.

ii.

State the Cauchy's criterion

"r"orro!:" of an infinite series.

Test the convergence of the series

,Ir(ffiw)

Note:

B.Tech. DEGREE EXAMINATION, NOVEMBER 2018 .

First Semester

ISlvIABlOtT - CALCULUS AND LINEAR ALGEBRA(For the candidates admitted during the academic

year* zeta --iota1

(D Part - A should be answered in OMR sheet within first 45 minutes and OMR sheet should be handed over to hall invigilator at the end of 45 minutes. (il) Part - B and Part - C should be answered in answer booklet.

Time: Three Hours

PART _Ae}xt=20Mayta)

Answer ALL questions

(tr4)1. The eigen values of A^2 , given $e = 1$ or $z \in I$ are

10 os)(A) 3,2,5 (B) 9,4,25 (c) 3,1,4 rnl g, r, re

2. The characteristic equation of the matrix $=(1 |$

,,

(A) $t^2 - 4a - s = 0$ (B) $7z'a_{41}_5 = s$ (c) $A^2 + 4.x + 5 = 0$ @) t , $*q_i_i_o$

(t 12\3. The nature of the quadratic form, whose matrix is $I t : f I i$,

1, t6)(A) Indefinite

(c) positive definite $E l L :: 'ffJ\#; * - "$

Max. Marks: 100

15NFTI8IUABlOrT

4' trth"canonical form of a quadratic form: $nis - fi * fi + 4rt$, then the signature of e is (A) 0 (B) 3 (c) 2

5. If $u = x^2 + 2y$, where $x = a^*$, $y = z$ at that $4'dt$

Ii] $z, t^2 + st + 1$ di $r, r, r(r^*r)$ (C) $4a^3 + 378 + 5, ty$ (D) $0oz' riltiSr >$

6' The stationary points of the function $3x^2 - y^2 + 13$ are (A) $(0, 0)$, $(-2, 0)$ (B) $(0, 0)$, $(2, 0)$ (c) $(1, 1)$, $(-2, 0)$ (D) $(1, 1)$, $(-2, 0)$

Page 1 of 4

FNFIISMABlOTTT

Page 4 of 4

** : f, S *

[Page 2]

7. The $x = r \cos \theta$, $y = r \sin \theta$, then the Jacobian of r, θ with respect to x, y is (A) r (B) r^2 (C) r^3 (D) r^4

(A) r

(C) $2r$

8. Maclaurin series for $\sin x$ is

$x - \frac{x^3}{6} + \dots$

(A) $1 + (x+y) - (x+y)^2 + \dots$

$3x$

$-\frac{(x-y)^3}{6},$

(C) $1 + (x-y) - \frac{(x-y)^2}{2} + \dots$

9. The solution of $(2x+4)y' = 0$ is

(A) $y = Ae^{2x} + Be^{-2x}$

(C) $y = Ae^{2x} + Be^{-2x}$

(A) $y = (7Ax + B)e^{-2x}$

(C) $y = Ae^{-x} + Be^{4x}$

15. The value of $\sin(\frac{\pi}{4})$ is

(A) $\frac{1}{\sqrt{2}}$

$\frac{1}{2}$

(C) $\frac{1}{2}$

(B) $\frac{1}{\sqrt{2}}$

$\frac{1}{\sqrt{2}}$

$(x - \sqrt{3})^2$

(B) $1 - (x-y)^2$

$-\frac{(x+y)^3}{6}$

$6p;$

$1 + (x+y) + \frac{(x+y)^2}{2} + \dots$

(B) $y = \arccos(x) + B \sin 2x$

(D) $y = \cos x + B \sin x$

When n is a positive integer, $F(n+1) =$

(A) $(n+1)!$

(C) $n!$

$\cos$

The series $\sum_{n=1}^{\infty} \frac{1}{n^p}$ is convergent if

$p > 1$

(A) $p > 1$

(C) $p > 1$

11. The series $\sum_{n=1}^{\infty} \frac{1}{n^p}$ is

(A) Convergent

(C) Divergent

series is convergent, if

(A) $p > 1$

(C) $p > 1$

(C) Divergent

(B) $n!$

(D) $(n-1)!$

(B) $p < 1$

(D) $p < 1$

@) Oscillating

@) Monotonic

(B) $\neq 0$

(D) < 1

@) conditionally convergent

(D) Absolutely convergent

16.

17.

19. As per D'Alembert's ratio test if $\sum_{n=1}^{\infty} a_n$ is a series of positive terms and $\lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = t$

then

10. The particular integral of the differential equation $(D^2 - 10D + 25)y = \cos x$ is

(A) $\frac{1}{2}(\cos x + 3\sin x)$

(B) $\frac{1}{2}(\cos x - 3\sin x)$

10'

(C) $\frac{1}{2}(\cos x - 3\sin x)$ (D) $\frac{1}{2}(\cos x + 3\sin x)$

11. The complementary function of $(D^2 + 4D + 4)y = e^{-2x}$ is

(A) ce^{-2x} (B) $ge^{-2x} + he^{-2x}$

^

(C) $Ce^{-2x} + B\sin 2x$ (D) $Ce^{-2x} + Be^{4x}$

12. The complementary function of $(D^2 + aD + 2)t = e^{2t}$ is

13. The radius of curvature of the curve $y = t^2$ at $t = 1$ is

(A) $\frac{1}{2}$, (B) $\frac{1}{4}$, (C) $\frac{1}{2}$

14. The radius of curvature of a circle at any point is same as its

(A) chord (B) radius (C) Diameter

PART-B (5x4=20 Marks)

Answer All the Questions

(-t) $\frac{1}{3}$

21. Use Cayley-Hamilton theorem to find A^{-1} , given that $A = \begin{pmatrix} 1 & 4 \\ 2 & 1 \end{pmatrix}$

22. If $\vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$

then $\nabla \cdot \vec{r} =$

Find the Jacobian of x, y, z with respect to

u, v, w where $x = uvw, y = \frac{u}{v}, z = \frac{w}{u}$

(2 - z)

24. Find the eigenvalues of the matrix $A = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$

u3

x_1, x_2, x_3 if $f(x_1, x_2, x_3) = x_1^2 + x_2^2 + x_3^2$

I

25.

26.

Solve the equation $(D^2 - 10D + 25)y = \sin 3x$

Find the envelope of the family of straight lines given by $x \cos \alpha + y \sin \alpha = a \sec \alpha$

15. If $f(x) = \sin x$, then $f'(x) =$

15. If $f(x) = \sin x$, then $f'(x) =$

Page 2 of 4

===== SOURCE_FILE: nexus/Calculus And Linear Algebra/PYQs/pyq txt/CT1 Papers 2025.txt =====

CATEGORY: PYQs

STATUS: ok

SRM Institute of Science and
Department of Mathematics

Academic Year: 2025-2026 - ODD

FT-II B2

Set- B

21MAB101T-Calculus and Linear Algebra

Date: 19.09.2025

Total Marks: 50 Time: 08.00-09.40 AM

Duration: 1 hour 40 Minutes

T34S10

Q. Part-A(3*8-24 Marks) Marks BL CO PO

No (Answer any THREE Questions)

1 Find the eigenvalues and eigenvectors of the matrix $\begin{pmatrix} 8 & 2 & 2 \\ 10 & -2 & -51 \end{pmatrix}$

$A = \begin{pmatrix} -2 & 2 \end{pmatrix}$

2 (a) Find the eigenvalues of the matrix

$\begin{pmatrix} -1 & 0 \\ 0 & y \end{pmatrix}$

$\begin{pmatrix} -3 & 0 \end{pmatrix}$ and hence find the eigenvalues of $\begin{pmatrix} 4 & 2 & 1 \end{pmatrix}$

$\begin{pmatrix} 4 & 2 \\ 1,2 \end{pmatrix}$

A by using the property.

(b) Discuss the nature of the Quadratic form $x^2 + 3x^2 + 6x + 2x^2 + 2x^2 + 4x^2$, without reducing to the canonical form.

3 Expand $x^2y + 3y^2 - 2$ in powers of $(x-1)$ and $(y+2)$ using Taylor's series up to second degree terms.

4 (a) If $z = xy + xy$, where $x = at$, $y = 2at$, then find $\frac{dz}{dt}$

without direct substitution. 2 2
dt 1,2

(b) If $u = \frac{x^2}{y}$ and $v = \frac{y^2}{x}$ then find $\frac{d^2u}{dy^2}$ 4 3 2 12
y

Part-B (2*13-26 Marks) Marks BL CO PO

No (Answer any TWO Questions)

5 Reduce the Quadratic form $3x^2 + 5xy + 3y^2$ into canonical form by an 13 4 1,2

orthogonal transformation and determine the signature, rank and nature of the Quadratic form.

6 Verify Cayley Hamilton theorem for the matrix $\begin{bmatrix} 1 & 3 & 1 \\ 3 & 1 & 1 \\ 1 & 1 & 2 \end{bmatrix}$ 13 3 1 1,2

and hence find A^{-1} .

$\begin{bmatrix} 1 & -1 & 2 \end{bmatrix}$

Find the volume of the largest rectangular parallelepiped that can be inscribed in the ellipsoid $x^2 + y^2 + z^2 = 1$. 13 4 2 1,2

Course Outcome (CO) and Bloom's level (BL) Coverage in Questions

Bloom's level

231

CO1 CO2 CO3 CO4 CO5 CO6

Evaluation Sheet

Name of the Student:

Register No.:

Part-A (3*8-24 Marks)

CO Allotted Awarded Total

No Marks Marks

8

2(a) 1 4

2(b) 1 4 Consolidated Marks:

3 2 8 CO Marks Scored

COI

4(a) 2 4

CO2

4(b) 2 4

Total

Part-B (2-13=26 Marks)

5 1 13

6 1 13

7 2

Signature of the Course Teacher

SRM Institute of Science and Technology

Department of Mathematics

Academic Year: 2025-2026 ODD

FT-II B2

Set-A

21MAB101T-Calculus and Linear Algebra

Time: 08.00-09.40 AM

Date: 19.09.2025 Duration: 1 hour 40 Minutes

Total Marks: 50

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Marks BL COPO

Part-A (3*8-24 Marks)

Q.

No (Answer any THREE Questions)

8 2 1 1,2

1 Find the eigenvalues and eigenvectors of the matrix

1 3\

2(a) If 2a nd 3 are the eigenvalues of the matrix

3 10 5\

A= -2 -3 find the third eigen value ofA and 2 1 1,2

3 5

hence find the eigenvalues of the matrix A- by using

property.

(b) Discuss the nature of the Quadratic form $3x^2 + 3x^4 + 2x^2 + 1,2$

$5x^2 - 2x^4 - 6x^3x - 6x^3x^1$ without reducing to the canonical form.

3 Expand esiny in powers of (x-1) and (y - ;) using 8 3 2 1,2

Taylor series up to second degree terms.

4 2 2 1,2

4 (a) If $x^e Y - 2ye^* = 1$, then find $\frac{dy}{dx}$

dx

(b) If $x = u^2 - v^2$ and $y = 2uv$, then find $O(9)$ 4 3 2 1,2

Part-B(2*13=26 Marks) Marks BL CO PO

Q. (Answer any TWØ Questions)

No

5Reduce the Quadratic form $22xf + x + x + 2x,x, - 13$ 4 1,2

$4x^3X, - 2x^3X$ into canonical form by an orthogonal transformation and determine the signature, rank and nature

of the Quadratic form.

6 Verify Cayley Hamilton theorem for the matrix

$\begin{bmatrix} 3 & 31 \\ 1 & 2 \end{bmatrix}$

$\begin{bmatrix} -1 & 4 \\ 3 & 1 \end{bmatrix}$ and hence find A^{-1} ,

$\begin{bmatrix} 3 & 1 \\ 1 & 2 \end{bmatrix}$

Find the dimensions of the rectangular box, open at the top of maximum capacity whose surface area is 432 sq. cm.

Course Outcome (CO) and Bloom's level (BL) Coverage in Questions

Blooms level

BLA,

239%

30

CO1 CO2 CO3 CO4 CO5 CO6

Evaluation Sheet

Name of the Student:

Register No.:

Part-A (3"8=24 Marks)

CO	Allotted	Awarded	Total
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No	Marks	Marks
----	-------	-------

8

Consolidated Marks:

2(a)	1	4
------	---	---

CO	Marks	Scored
----	-------	--------

2(b)	1	4
------	---	---

CO1

3 2 CO2

4(a) 2 4 Total

4(b) 2

Part-B (2*13-26 Marks)

5 13

6 13

Signature of the Course Teacher

2 13

===== SOURCE_FILE: nexus/Calculus And Linear Algebra/PYQs/pyq txt/MCQ Unit 1 to 5.txt =====

CATEGORY: PYQs

STATUS: ok

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Multiple Choice Questions

Calculus and linear algebra (SRM Institute of Science and Technology)

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Unit – I: Matrices

PART A

MULTIPLE CHOICE QUESTIONS

1. The matrix of the quadratic form $x^2 + 5x_2 + x_2$

$1 \ 2 \ 3 \ 2x_1x_2 \ 2x_2x_3 \ 6x_3x_1$ is

$\begin{pmatrix} 1 & 1 & 3 \\ 1 & 2 & 3 \\ 1 & 4 & 4 \end{pmatrix}$ $\begin{pmatrix} 1 & 4 & 3 \\ 1 & 2 & 3 \\ 1 & 4 & 4 \end{pmatrix}$

$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 4 \\ 3 & 4 & 1 \end{pmatrix}$ $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 4 \\ 3 & 4 & 1 \end{pmatrix}$

(a) $\begin{pmatrix} 1 & 5 & 1 \\ 1 & 5 & 2 \\ 4 & 5 & 3 \end{pmatrix}$ (b) $\begin{pmatrix} 1 & 5 & 2 \\ 4 & 5 & 3 \\ 4 & 5 & 4 \end{pmatrix}$

$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 4 \\ 3 & 4 & 1 \end{pmatrix}$ $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 4 \\ 3 & 4 & 1 \end{pmatrix}$

$\begin{pmatrix} 3 & 1 & 1 \\ 3 & 2 & 1 \\ 4 & 3 & 1 \end{pmatrix}$ $\begin{pmatrix} 3 & 1 & 1 \\ 3 & 2 & 1 \\ 4 & 3 & 1 \end{pmatrix}$

2. The number of positive terms in the canonical form is called

(a) Signature (b) Index (c) Quadratic form (d) Positive definite

3. A homogeneous polynomial of second degree in any number of variables is

(a) Canonical form (b) Quadratic form (c) Orthogonal (d) Diagonal form

Q3 2 4Q

4. Find the eigen values of A^2 if $A = \begin{bmatrix} 2 & 6 \\ 0 & 2 \end{bmatrix}$

Q40 0 5Q

(a) 6, 4, 10 (b) 9, 4, 25 (c) 9, 2, 5 (d) 3, 2, 5

Q2 1 1Q

5. Find the sum and product of the eigen values of $A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$

Q40 0 1Q

(a) 5, 3 (b) 3, 5 (c) 2, 1 (d) 0, 1

6. The eigen values of an orthogonal matrix have the absolute value _____

(a) 0 (b) 1 (c) 2 (d) ± 1

7. All the eigen values of a symmetric matrix with real elements are

(a) Distinct (b) Real (c) Equal (d) Conjugate complex numbers

8. Find the nature of the quadratic form $2x^2 + 3y^2 + 2z^2 + 2xy$

(a) Positive definite (b) Negative definite (c) Positive semi-definite (d) Indefinite

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Q 6 1 7Q

9. Write the Q.F. defined by the matrix $\begin{bmatrix} 1 & 2 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

$A = \begin{bmatrix} 1 & 2 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

Q

$$\begin{bmatrix} 7 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} \end{bmatrix}$$

(a) $\begin{bmatrix} 6 \times 2 & 2 \times 2 & x^2 & 2x & x^2 & 14x & x \end{bmatrix}$ (b) $\begin{bmatrix} 6 \times 2 & x^2 & 6 \times 2 \\ 1 & 2 & 3 & 1 & 2 & 1 & 3 & 1 & 2 & 3 \end{bmatrix}$ $\begin{bmatrix} x^2 & 7x^2 & 1x^3 \end{bmatrix}$

(c) $\begin{bmatrix} 6 \times 2 & 2 \times 2 & x^2 & 2x & x^2 & 14x & x \end{bmatrix}$ (d) $\begin{bmatrix} 6 \times 2 & x^2 & 6 \times 2 \\ 1 & 2 & 3 & 1 & 2 & 1 & 3 & 1 & 2 & 3 \end{bmatrix}$ $\begin{bmatrix} x^2 & 14x^2 & 1x^3 \end{bmatrix}$

$$\begin{bmatrix} 2 & 1 \end{bmatrix}$$

10. Find the eigen values of the matrix $\begin{bmatrix} \end{bmatrix}$

$$\begin{bmatrix} 1 & 2 \end{bmatrix}$$

(a) 1, 3 (b) 3, 1 (c) 2, 1 (d) 1, 2

11. Find the eigen values of $\begin{bmatrix} 10 & 1 & 2 \end{bmatrix}$
A if $A = \begin{bmatrix} \end{bmatrix}$

$$\begin{bmatrix} 0 & 3 \end{bmatrix}$$

(a) 1,310 (b) 3,1 (c) 32,110 (d) 0,2

12. If the sum of two eigen values and trace of a 3 x 3 matrix A are equal, find the value of A

(a) 0 (b) 1 (c) -1 (d) 2

$$\begin{bmatrix} 1 & 2 & 3 \end{bmatrix}$$

13. Find the characteristic equation of the matrix $A = \begin{bmatrix} \end{bmatrix}$

$$\begin{bmatrix} \end{bmatrix}$$

$$\begin{bmatrix} 3 & 1 & 1 \end{bmatrix}$$

$$\begin{pmatrix} \end{pmatrix} \begin{bmatrix} \end{bmatrix}$$

a $\begin{bmatrix} 2 & 18 & 40 \end{bmatrix}$ (b) $\begin{bmatrix} 3 & 2 & 18 \end{bmatrix} 40$

(c) $\begin{bmatrix} 3 & 2 & 18 \end{bmatrix} 40$ (d) $\begin{bmatrix} 3 & 2 & 18 \end{bmatrix} 40$

14. Find the nature of the quadratic form $x^2 + 2x^2 + 3x^2$
 $\begin{bmatrix} 1 & 2 & 3 \end{bmatrix}$ $\begin{bmatrix} 2x^2 & 2x^2 & 2x^2 \end{bmatrix}$ $\begin{bmatrix} 2x^2 & 2x^2 & 2x^2 \end{bmatrix}$

(a) Indefinite (b) Positive definite (c) Negative definite (d) Positive semidefinite

$$\begin{bmatrix} 2 & 2 & 0 \\ 2 & 2 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

15. Find the eigen values of $A = \begin{bmatrix} 2 & 1 & 1 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{bmatrix}$

$$\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{bmatrix}$$

(a) 1, 3, 4 (b) 1, 3, 4 (c) 1, 3, 4 (d) 1, 3, 4

16. The matrix of the quadratic form $2x^2 + 4xy + 2y^2$ is

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$$\begin{bmatrix} 1 & 1/2 \\ 1/2 & 1 \end{bmatrix}$$

(a) $\begin{bmatrix} 1 & 1/2 \\ 1/2 & 1 \end{bmatrix}$ (b) $\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$ (c) $\begin{bmatrix} 1 & 1/2 \\ 1/2 & 1 \end{bmatrix}$ (d) $\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$

$$\begin{bmatrix} 1/2 & 0 \\ 0 & 1/2 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}$$

17. Two eigen values of the matrix $\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$

$A = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$ are 1 and 2. Find the third eigen value.

$$\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$$

(a) 3 (b) b (c) 2 (d) 1

18. Two of the eigen values of 3×3 matrix A are 2, 1 and $|A| = 12$. Find the third eigen value

(a) 6 (b) 3 (c) 2 (d) 1

19. If A is an orthogonal matrix then

(a) $|A| = 0$ (b) A is singular (c) $A^2 = I$ (d) $A^T = A^{-1}$

$$\begin{bmatrix} 4 & 6 & 6 \\ 6 & 4 & 6 \\ 6 & 6 & 4 \end{bmatrix}$$

20. Two eigen values of $A = \begin{bmatrix} 1 & 3 & 2 \\ 3 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix}$

are equal and they are double the third. Find them.

Q15 Q22

(a) 1, 2, 2 (b) 2, 1, 1 (c) 2, 0, 1 (d) 1, 2, 3

Q1 Q22

21. Find the inverse of the eigen values of the matrix if $A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$

Q5 4 Q

Q

(a) $1, 1/6$ (b) $1, 1/6$ (c) $1, 1/6$ (d) $1, 1/6$

22. Find rank and index of the QF whose canonical form is $3y^2 - 3y^2$
2 3

(a) 2, 1 (b) 1, 2 (c) 0, 1 (d) 0, 2

23. Find signature of the QF whose canonical form is $2y^2 - y^2 - y^2$
1 2 3 ,

(a) 1 (b) -1 (c) 0 (d) 6

24. The eigen vectors corresponding to the distinct eigen values of a real symmetric matrix are

(a) imaginary (b) non-orthogonal (c) real (d) orthogonal

Q 0 1 Q

25. Find the characteristic equation of the matrix $A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$

Q2 Q322

(a) $2 - 3\lambda^2 - \lambda = 0$ (b) $2 - 3\lambda^2 - 2 - \lambda = 0$ (c) $2 - 3\lambda^2 - 3\lambda = 0$ (d) $2 - 6\lambda^2 - 3\lambda = 0$

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UNIT-II-FUNCTIONS OF SEVERAL VARIABLES

1. If $Z = x^2 + 3y^2$ then what is ?

(i) $2y+3x$ (ii) $3y$ (iii) $2x+3y$ (iv) $2x$

2. $u =$ is homogeneous function of degree

- (i) 2 (ii) 3 (iii) 1 (iv) 4

3. If $u = a + 2hxy + b$ then using Euler's theorem find $x + y = ?$

- (i) u (ii) $2u$ (iii) $3u$ (iv) $n(n-1)$

4. If $f(x, y) =$ then what is $(1, 1) ?$

- (i) $-e$ (ii) $(iii)e$ (iv) $-$

5. if $z = \log(x + y)$ then what is $x + y = ?$

- (i) 1 (ii) $(iii) 2$ (iv)

6. If $f(x, y)$ is an implicit function then $= ?$

- (i) (ii) (iii) (iv)

7. If $f(x, y) = ?$

- (i) 1 (ii) -1 (iii) 0 (iv) 2

8. If $f(x, y) = \cos x \cos y$ then $(0, 0) = ?$

- (i) 1 (ii) 0 (iii) -1 (iv)

9. If $f(x, y) =$ then $(1, 1)$ is

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- (i) (ii) $\frac{1}{2}$ (iii) $-\frac{1}{2}$ (iv) 0

10. If $r^2 = at^2 + b$ then the point is

- (i) Maximum point (ii) minimum point (iii) saddle point (iv) none of these

11. The stationary points of are

- (i) $(-3, 0)$ (ii) $(0, 3)$ (iii) $(0, -3)$ (iv) $(3, 0)$

12. If $x =$ and $y = 2uv$ then J is

- (i) (ii) 2 (iii) 4 (iv) 4

13. If $x = r$ and $y = r$ Then what is $=$?

- (i) (ii) r (iii) $2r$ (iv) 0

14. If $v = +$ then is

- (i) $1+$ (ii) (iii) (iv) $1+$

15. u and v are functionally dependent if their jacobian value is

- (i) zero (ii) one (iii) non-zero (iv) greater than zero

16. if $and =$ then $=?$

- (i) 0 (ii) 1 (iii) -1 (iv) 2

17. The stationary points of $f(x, y) =$ are

- (i) (ii)

18. The point $(0, 0)$ for $f(x, y) =$ is

- (i) a maximum point (ii) a minimum point (iii) a saddle point (iv) none of these

19. If $f(x, y) =$ where $x = r \cos$ and $y = r \sin$ then is

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- (i) r (ii) (iii) 1 (iv) 0

20. If $f(x, y) = y + \sin y +$ then $(1,)$ is

- (i) $2 - e$ (ii) 2π (iii) $2\pi + e$ (iv) 0

21. $u =$ is homogeneous function of degree

- (ii) 1 (iii) 2 (iv) 3

22. If $u =$ then $x + y = ?$

(i) $\sin u$ (ii) $\cos u$ (iii) $\sin 2u$ (iv) $\tan u$

23. If $u =$, $v =$ then $= ?$

(i) -3 (ii) 3 (iii) $-$ (iv)

, $y = r$ $z = z$ then $= ?$

(i) $2r$ (ii) (iii) (iv) r

25. If $u =$ and $v = x + y$ then $= ?$

(i) $2x$ (ii) $2x + 2$ (iii) $2y - 2$ (iv) $2x - y$

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ANSWERS

1. (iii) $2x + 3y$

2. (iii) 1

3. (ii) $2u$

4. (iii) e

5. (iii) 2

6.i)

7. (iii) 0

8. (iii) - 1

9. (ii) $\frac{1}{2}$

10. (iii) saddle point

11. (i) (-3 , 0)

12. (iii) 4(

13. (ii) r

14. (iii)

15. (i) zero

16. (ii) 1

17. (ii)

18. (iii) a saddle point

1290.. ((iivii)) $20\pi + e$

21.

22. (iii) $\sin 2u$

23. (iii) -

24. (iv) r

25. (ii) $2x+2$

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Unit-III Ordinary Differential Equations

Multiple Choice Questions

1. Which of the following is the general solution to $\frac{d^2y}{dx^2} + 3 \frac{dy}{dx} - 10y = 0$

(a) $y = Ae^{2x} + Be^{-5x}$ (b) $y = Ae^{-2x} + Be^{5x}$ (c) $y = Ae^{-2x} + Be^{-5x}$ (d) $y = Ae^{2x} + Be^{5x}$

2. Solution of $(D^2 + 4)y = 0$ is
 $\sqrt{\quad} \sqrt{\quad}$

(a) $y = A \cos 2x + B \sin 2x$ (b) $y = Ae^{2x} + Be^{-2x}$ (c) $y = A \cos 2x + B \sin 2x$
(d) $y = (Ax+B)e^{2x}$

3. The P.I of $(D^2 + 4)y = \sin 2x$ is

(a) $-\frac{x}{4} \cos 2x$ (b) $\frac{x}{4} \cos 2x$ (c) $\frac{x}{2} \cos 2x$ (d) $-\frac{x}{2} \cos 2x$

4. The equation $(a_0 \frac{d^2}{dx^2} + a_1 \frac{d}{dx} + a_2) y = Q(x)$ is called, where $a_0, a_1, a_2 \in \mathbb{C}$

(a) Cauchy's equation (b) Legendre's equation (c) Taylor's equation (d) Clairaut's equation

,

5. Use the transformation $z = \log x$, convert the D.E $x^2 \frac{d^2y}{dx^2} -$

$y -$

$xy + y = x^2$ to an equation with
constant coefficients

(a) $(\theta^2 - 2\theta + 1)y = e^{2z}$ (b) $(\theta^2 - 2\theta + 1)y = e^z$ (c) $(\theta^2 + 2\theta + 1)y = e^{2z}$

(d) $(\theta^2 + 2\theta + 1)y = e^z$

6. The solution of $(D^2 + 2D + 1)y = 7$ is

- (a) $y = (Ax + B)e^{-x} + 7$ (b) $y = (Ax + B)e^{-x} - 7$ (c) $y = (Ax + B)e^x + 7$
(d) $y = (Ax + B)e^x - 7$

7. The P.I of $(D - 1)^2y = e^x \sin x$ is

- (a) $-e^x \cos x$ (b) $e^x \cos x$ (c) $e^x \sin x$ (d) $-e^x \sin x$

8. The P.I of $(D - 1)^2y = x$ is

- (a) $2 - x$ (b) $x + 2$ (c) x^2 (d) $-x^2$

9. If $1 \pm 2i$ are the roots of A.E of a differential equation $f(D)y = 0$ then the general solution is

- (a) $e^{-2x} (A \cos x - B \sin x)$ (b) $Ae^x + Be^{-2x}$ (c) $e^x (A \cos 2x + B \sin 2x)$ (d) $Ae^t + Be^{2x}$

,

10. Convert the equation $(5 + 2x)^2$

' _ '

$y^6(5 + 2x)y + 8y = 0$ to an equation with constant coefficient by using the transformation $z = \log(5 + 2x)$

- (a) $(\theta^2 + 4\theta + 2)y = 0$ (b) $(\theta^2 - 4\theta + 2)y = 0$ (c) $(\theta^2 + 4\theta + 4)y = 0$ (d) $(\theta^2 + 4\theta - 2)y = 0$

11. The P. I of $(D^2 + 4)y = \sinh 2x$ is

- (a) $y_p = \sinh 2x$ (b) $y_p =$

$\sinh 2x$ (c) $y = -\sinh 2x$ (d) $y = \sinh 2x$
 $4 p 8 p = 8 4$

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12. The P.I of $(D^2 + 6D + 5)y = e^{-x}$ is

(a) $y_p =$
 $x e^{-x}$

(b) $y_p =$
 $x e^{-x} - x$

(c) $y = e$ (d) $y = e^{-x}$

$4 \frac{d^2 p}{dx^2} + p = 4$

13. The solution of $(D^2 - 2aD + a^2)y = 0$ is

(a) $Ae^{ax} + Be^{bx}$ (b) $Ae^{ax} + Be^{-ax}$ (c) $(Ax + B)e^{ax}$ (d) $(Ax + B)e^{-ax}$

14. The P.I of $(D^2 + 16)y = \cos 4x$ is

(a) $x \sin 2x$ (b) $x \sin 4x$ (c) $x \cos 2x$ (d) $x \cos 4x$
 $\frac{1}{2} \frac{1}{8} \frac{1}{2} \frac{1}{8}$

15. The C.F of $D^2 y + y = \operatorname{cosec} x$ is

(a) $Ae^{ax} + Be^{bx}$ (b) $A \cos x + B \sin x$ (c) $(Ax + B)e^{ax}$ (d) $(Ax + B)e^{-ax}$

$\frac{1}{x} - \frac{1}{x^2}$

16. If $y_1 = \cos ax$, $y_2 = \sin ax$ then the value of $y_1 y_2$ is

1

(a) $-a$ (b) 0 (c) 1 (d) a

17. Solve $(D^2 + 1)y = 0$ given $y(0) = 0$, $y'(0) = 1$

(a) $y = \sin x$ (b) $y = \cos x$ (c) $y = A \cos x + B \sin x$ (d) $y = 0$

18. The P.I of $(D - 2)^2 y = e^{2x}$ is

(a) x^2

e^{2x} (b) $x e^{2x}$ (c) $x e^{-2x}$ (d) x^2

e^{-2x}

2 4 2 2

19. The P.I of $(D^2 + 4)y = \sin(2x + 5)$ is

(a) $-x \sin(2x+5)$ (b) $x \sin(2x+5)$ (c) $-x \cos(2x+5)$ (d) $x \cos(2x+5)$

2 4 4 2

20. Solve $(x^2 D^2 + xD + 1) y = 0$ is

(a) $Ae^{az} + Be^{bz}$ (b) $A \cos z + B \sin z$ (c) $(Az + B)e^{az}$ (d) $(Az + B)e^{-az}$

21. The roots of the auxiliary equation $(m^2 - 4) = 0$ are

✓

(a) ± 2 (b) $\pm 2i$ (c) ± 2 (d) $1 \pm 2i$

22. The solution of $(x^2 D^2 - 7xD + 12) y = 0$ is

(a) $Ae^{-2z} + Be^{6z}$ (b) $Ae^{2z} + Be^{-6z}$ (c) $Ae^{2z} + Be^{6z}$ (d) $Ae^{-2z} + Be^{-6z}$

' _ '

23. If $y_1 = \cos x$, $y_2 = \sin x$ then the value of $y_1 y_2$ is

2 2y s

1

(a) -1 (b) 0 (c) 1 (d) 1

2

24. If three roots of the auxiliary equation become equal to the real number a , then the corresponding C.F is

(a) $(Ax^2 + Bx + C) e^{ax}$ (b) $Ae^{ax} + Be^{ax} + Ce^{ax}$ (c) $Ae^{ax} + (B \cos ax + C \sin ax)$ (d) a

25. The values of e^{ax}

D-a

(a) xe^{ax} (b) e^{ax} (c) $x^2 e^{ax}$ (d) x^2

e^{ax}

2

Answers:

1. a 2. a 3. a 4. a 5. a 6. a 7. d 8. b 9. c 10. b 11. a
12. a 13. c 14. b 15. b 16. d 17. a 18. a 19. c 20. b 21. b
22. c 23. c 24. a 25. a

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SRM UNIVERSITY

MA1001- CALCULUS AND SOLID GEOMETRY

Unit-IV Geometrical Applications of Differential Calculus

Multiple Choice Questions

1. If the radius of curvature and curvature of a curve at any point are ρ and κ respectively, then

(a) $\rho = -1$

κ (b) $\rho = \kappa$ (c) $\rho = -\kappa$ (d) $\rho = 1$

κ

2. The locus of center of curvature is called

(a) Involute (b) Evolute (c) Radius of curvature (d) Envelope

3. The envelope of the family of curves $A\alpha^2 + B\alpha + C = 0$ (α is parameter) is

(a) $B^2 + 4AC = 0$ (b) B^2

$-AC = 0$ (c) $B^2 + AC = 0$ (d) B^2

$- 4AC = 0$

4. The curvature of the straight line is

(a) 1 (b) 2 (c) -1 (d) 0

5. The radius of curvature of the curve $y = 4 \sin x$ at $x = \pi$ is

2

(a) 1 (b) -1 (c) 1 (d) -1

2 2 4 4

6. The envelope of $ty - x = at^2$, t is the parameter is

(a) $x^2 = 4ay$ (b) $y^2 = 4ax$ (c) $x^2 + y^2 = 1$ (d) $x^2 - y^2 = 1$

7. The curvature at any point of the circle is equal to --- of its radius

(a) Square (b) Same (c) Reciprocal (d) constant

8. What is the radius of curvature at (4, 3) on the curve $x^2 + y^2 = 25$

(a) 5 (b) -5 (c) 25 (d) -25

9. What is the curvature of a circle of radius 3

(a) 3 (b) -3 (c) 1 (d) -1

3 3

10. Find the envelope of the curve $y = mx + a$
m where m is a parameter

(a) $y^2 - 4ax = 0$ (b) $y^2 + 4ax = 0$ (c) $x^2 + y^2 = 1$ (d) $xy = c^2$

11. The radius of curvature of $y = e^x$ at $x = 0$ is

$\sqrt{2}$

(a) $\sqrt{2}$ (b) 2

$\sqrt{2}$ (c) 2 (d) 1

$\sqrt{2}$

2 2

12. The radius of curvature of the curve $y = \log \sec x$ at any point of it is

(a) $\sec x$ (b) $\tan x$ (c) $\cot x$ (d) $\operatorname{cosec} x$

13. In an ellipse the radius of curvature at the end of which axis is equal to the semi latus rectum of the ellipse

(a) Minor (b) Major (c) Vertical (d) Horizontal

14. The radius of curvature of the curve $x = t^2$, $y = t$ at $t = 1$ is
 $\sqrt{5}$

5 5 5 $\sqrt{5}$

(a) $\frac{5}{2}$ (b) $\frac{5}{2}$ (c) $\frac{5}{2}$ (d) $\frac{5}{2}$

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15. Evolute of a curve is the envelope of ---- of that curve

(a) Tangent (b) Normal (c) Parallel (d) Locus

16. The evolute of the cycloid $x = a(\theta - \sin \theta)$, $y = a(1 - \cos \theta)$ is

(a) Astroid (b) Parabola (c) Cycloid (d) Circle

17. A curve which touches each member of a family of the curves is called --- of that family

(a) Evolute (b) Envelope (c) Circle of curvature (d) Radius of curvature

18. The envelope of family of lines $y = mx + am^2$ (where m is the parameter) is

(a) $x^2 + 2ay = 0$ (b) $x^2 + 4ay = 0$ (c) $y^2 + 2ax = 0$ (d) $y^2 + 4ax = 0$

19. The envelope of the family of lines $xt + yt = 2c$, t being the parameter is

(a) $x^2 + y^2 = c^2$ (b) $xy = c^2$ (c) $x^2y^2 = c^2$ (d) $x^2 - y^2 = c^2$

20. The radius of curvature at any point on the curve $r = e^\theta$ is
 $\sqrt{2}$

(a) $\frac{2}{r}$ (b) $\frac{2}{r}$

$$\sqrt{(c)^2 + (d)^2}$$

21. The radius of curvature in Cartesian coordinates is
 $\frac{1}{\rho} = \frac{1}{r^2} \left(\frac{dy}{dx} \right)^2 + 1$

$$\frac{1}{\rho} = \frac{1}{r^2} \left(\frac{dy}{dx} \right)^2 + 1$$

$$\rho = \frac{1}{\sqrt{\left(\frac{dy}{dx} \right)^2 + 1}}$$

$$\rho = \frac{1}{\sqrt{\left(\frac{dy}{dx} \right)^2 + 1}}$$

$$\frac{1}{\rho} = \frac{1}{r^2} \left(\frac{dy}{dx} \right)^2 + 1$$

$$2$$

$$y^1$$

22. The radius of curvature in polar coordinates is
 $\frac{1}{\rho} = \frac{1}{r^2} \left(\frac{dr}{d\theta} \right)^2 + 1$

$$\frac{1}{\rho} = \frac{1}{r^2} \left(\frac{dr}{d\theta} \right)^2 + 1$$

$$\frac{1}{\rho} = \frac{1}{r^2} \left(\frac{dr}{d\theta} \right)^2 + 1$$

$$\rho = \frac{1}{\sqrt{\left(\frac{dr}{d\theta} \right)^2 + 1}}$$

$$\frac{1}{\rho} = \frac{1}{r^2} \left(\frac{dr}{d\theta} \right)^2 + 1$$

$$\frac{1}{\rho} = \frac{1}{r^2} \left(\frac{dr}{d\theta} \right)^2 + 1$$

$$\frac{1}{\rho} = \frac{1}{r^2} \left(\frac{dr}{d\theta} \right)^2 + 1$$

$$\frac{1}{\rho} = \frac{1}{r^2} \left(\frac{dr}{d\theta} \right)^2 + 1$$

$$\frac{1}{\rho} = \frac{1}{r^2} \left(\frac{dr}{d\theta} \right)^2 + 1$$

$$= -r r' + 2(r')^2$$

$$= (r')^3/2$$

,

$$r^2 + (r')^2$$

(d) $\rho =$

,

$$r^2$$

,

$$= -r r' + 2(r')$$

$$r')^2$$

23. The radius of curvature in parametric coordinates is

$$(r')^3/2 (r')^3/2 (r')^3/2$$

, , , , , ,

$$(x')^2 + (y')^2 (x')^2 + (y')^2 (x')^2 - (y')^2$$

(a) $\rho =$ (b) $\rho =$ (c) $\rho =$

, , , , , , , ,

$$x y x y + '' ' '' ' ''$$

$$= -y x y x x y - y x$$

$$(r')^3/2$$

, ,

$$(x')^2 - (y')^2$$

(d) $\rho =$

, , , ,

$$x y + '$$

$$y x$$

24. The equation of circle of curvature at any point (x, y) with center of curvature x, y and with

radius of curvature ρ is

$$(a) (x+x)^2 + (y+y)^2 = \rho^2 \quad (b) (x-x)^2 + (y-y)^2 = \rho^2 \quad (c) (x-x)^2 - (y+y)^2 = \rho^2$$

$$(d) (x+x)^2 + (y+y)^2 = \rho$$

Answers:

1. d 2. b 3. d 4. d 5. c 6. b 7. c 8. a 9. c 10. b 11. a 12. a
13. b 14. a 15. b 16. c 17. b 18. b 19. b 20. d 21. a 22. d 23.
a 24. b

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SRM INSTITUTE OF SCIENCE & TECHNOLOGY

FACULTY OF ENGINEERING AND TECHNOLOGY

DEPARTMENT OF MATHEMATICS

Unit -IV Geometrical Applications of Differential Calculus

(Beta ,Gamma Functions)

Multiple Choice Questions

1. The value of $\beta(4,4)$ is -----

$$\frac{36}{6!} \frac{4!4!}{3!}$$

(a). (b). (c). (d).

$$\frac{7!}{7!} \frac{8!}{7!}$$

2. The value of $\beta\left(\frac{5}{2}, \frac{3}{2}\right)$

is -----

$$\frac{\pi}{2} \frac{2}{2}$$

$$\frac{\pi}{2} \frac{\pi}{2} \frac{\pi}{2} \frac{\pi}{2}$$

(a). (b). (c). (d).

8 8 16 16

3. $\varphi(\varphi, \varphi)$ is equal to -----

$$m!n! \quad m!n! \quad \Gamma_m \Gamma_n \quad \Gamma_m \Gamma_n$$

(a). (b). (c). (d).

$$(m+n)! \quad (m-n)! \quad \Gamma(m+n) \quad \Gamma(m-n)$$

1

4. The value of $\Gamma\left(\frac{1}{2}\right)\Gamma\left(\frac{1}{2}\right)$ is -----

2

(a). π (b). π^2

(c). π (d). 2π

5. $\Gamma(n) \Gamma(1-n)$ is equal to -----

$$\propto 1-n$$

X

(a). $\int_0^1 dx$ (b). $\Gamma(1) \beta(n, 1 - n)$ (c). $\Gamma(1) \beta(1 - n, 1 - n)$ (d). $\Gamma(1) \beta(1 - n, n)$

 $1+x$

0

 π

2

p q

6. $\sin \theta \cos \theta d\theta$ is equal to -----

0

1

(a) $\beta \in p+1 \ q+1 \ 1$

$$\beta = p^{-1} q^{-1} 1 1$$

(b). α , β (c). $\beta \wedge p \vee q \wedge \beta \wedge 1 \vee 1 \wedge$

□, □ (d). □, □

2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2

1

$$4 \cdot 1$$

$$[]^3$$

7. $\int x \log(x) dx$ is equal to -----
 x

$$0$$

$$6 \cdot 6 \cdot 6! \cdot 5!$$

(a). (b). (c). (d).

$$525 \cdot 625 \cdot 5! \cdot 6!$$

$$\pi$$

$$2$$

8. The value of $\int_0^{\pi} \cot \theta d\theta$ is -----
 0

$$\pi \cdot \pi \cdot \pi \cdot \pi$$

(a). (b). (c). (d).

$$2 \cdot 2 \cdot 2 \cdot 2$$

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UNIT V

Sequence and Series

$$\sum_{n=1}^{\infty} \frac{1}{n^p}$$

$$\sum_{n=1}^{\infty} \frac{1}{n^p}$$

1. The series is convergent if

$$p$$

$$n \geq 1$$

$$(a)p=1 \quad (b)p=0 \quad (c)p>1 \quad (d)p<1$$

$$\sum_{n=1}^{\infty} \frac{1}{n^p}$$

$$\sum_{n=1}^{\infty} \frac{1}{n^p}$$

2. The series is divergent if

$\sum_{n=1}^{\infty} \frac{1}{n^p}$

(a) $p > 1$ (b) $p = 0$ (c) $p \leq 1$ (d) $p < 1$
1

3. If $\sum u_n$ is a series of positive term such that $\lim_{n \rightarrow \infty} \frac{u_{n+1}}{u_n} = l$ where $l > 1$, then the series $\sum u_n$ is

(a) Convergent (b) Divergent (c) Oscillating (d) Monotonic
1

4. If $\sum u_n$ is a series of positive term such that $\lim_{n \rightarrow \infty} \frac{u_{n+1}}{u_n} = l$ where $l < 1$, then the series is convergent

if

(a) $l < 1$ (b) $l = 0$ (c) $l > 1$ (d) $l = 1$
2

2

1
5. The series is

$\sum_{n=1}^{\infty} \frac{1}{n!}$
(a) Convergent (b) Divergent (c) Oscillating (d) Monotonic

1 1 1
6. The series $1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \dots$ to ∞ is

2 3 4
(a) Convergent (b) Divergent (c) Oscillating (d) Monotonic

u
7. By D'Alembert's Ratio test $\lim_{n \rightarrow \infty} \left| \frac{u_{n+1}}{u_n} \right| = l$, the series is convergent when

$\lim_{n \rightarrow \infty} \frac{u_{n+1}}{u_n} = l$
(a) $l < 1$ (b) $l = 0$ (c) $l > 1$ (d) $l = 1$

2 u
8. By Raabe's test $\lim_{n \rightarrow \infty} n \left(\frac{u_n}{u_{n+1}} - 1 \right) = l$

$m, n, n-1, 1$)

$\sum_{n=1}^{\infty} \frac{1}{n^l}$, the series is divergent when

$n \geq 1$ un $\geq$

(a) $l < 1$ (b) $l = 0$ (c) $l > 1$ (d) $l = 1$

$\sum_{n=1}^{\infty} \frac{1}{n^2}$

$\sum_{n=1}^{\infty} 2^n$

9. The series is

$\sum_{n=1}^{\infty} n!$

(a) Convergent (b) Divergent (c) Oscillating (d) Monotonic

10. A series $\sum_{n=1}^{\infty} u_n$ is said absolutely convergent if the series

(a) $\sum_{n=1}^{\infty} u_n$ is convergent (c) $\sum_{n=1}^{\infty} u_n$ is divergent

(c) $\sum_{n=1}^{\infty} u_n$ is convergent (b) $\sum_{n=1}^{\infty} u_n$ is divergent

11. A series $\sum_{n=1}^{\infty} u_n$ is said conditionally convergent if the series

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(a) $\sum_{n=1}^{\infty} u_n$ is convergent (b)

$\sum_{n=1}^{\infty} u_n$ is divergent &

$\sum_{n=1}^{\infty} u_n$ is convergent

$\sum_{n=1}^{\infty} u_n$

(c) $\sum_{n=1}^{\infty} u_n$ is convergent &

$\sum_{n=1}^{\infty} u_n$ is divergent (d)

$\sum_{n=1}^{\infty} u_n$ is divergent

$\sum_{n=1}^{\infty} u_n$

$\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \dots$

12. The series $\sum_{n=1}^{\infty} \frac{1}{n^2}$ is

$\frac{2}{3}, \frac{2}{5}, \frac{2}{7}, \dots$

(a) Convergent (b) Divergent (c) Conditionally convergent (d) absolutely convergent

1 1 1

13. The series $1 - \frac{1}{2} + \frac{1}{4} - \frac{1}{8} + \dots$ is

2 3 4

(a) Convergent (b) Divergent (c) Conditionally convergent (d) absolutely convergent

$\frac{1}{n}$

14. The series is

$n \log n$

(a) Conditionally convergent (b) absolutely convergent (c) Convergent (d) Divergent

15. An absolutely convergent series is

(a) Conditionally convergent (b) absolutely convergent (c) Convergent (d) Divergent

$\frac{1}{n}$

16. The series is

3^n

(a) Conditionally convergent (b) absolutely convergent (c) Convergent (d) Divergent

$\frac{1}{n}$

17. The series is

$(\log n)^n$

(a) Convergent (b) Conditionally convergent (c) absolutely convergent (d) Divergent

ANSWERS

1. d 11. c

2. c 12. d

3. b 13. c

4. a 14. d

5. a 15. c

6. b 16. c

7. a 17. a

8. a

9. b

10. a

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CATEGORY: PYQs

STATUS: ok

Reg. No

B.Tech./M.Tech (Integrated) DEGREE EXAMINATION, JANUARY 2023

First Semester

21MABI01T-CALCULUS AND LINEAR ALGEBRA

(For the candidates admitted from the academic year 2022-2023)

Note:

Part A should be answered in OMR sheet within first 40 minutes and OMR sheet should be handed

over to hall invigilator at the end of 44 minute.

() Part-B and Part - C should be answered in answer booklet.

Time: 3 Hours Max. Marks: 75

Marka M Co o

PART A (20 x 1 20Marks)

Answer ALL Questions

1. The number of positive terms in the canonical form is called

- (A) Signature (B) Index
(C) Quadratic form (D) Positive definite

2. $\begin{pmatrix} 1 & 2 & 4 \\ 0 & 0 & 5 \end{pmatrix}$

Find the eigen values of A if $A = \begin{pmatrix} 0 & 2 & 6 \\ 0 & 0 & 5 \end{pmatrix}$

$\begin{pmatrix} 0 & 0 & 5 \end{pmatrix}$

- (A) 1,4,25 (B) 2,4, 20
(C) 4,4, 25 (D) 1,2,25

3. If A is an orthogonal matrix then A^j is I

(A) $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ (B) $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$

(C) $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ (D) $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$

4. Find the sum and product of the eigen values of the matrix $\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$

$\begin{pmatrix} 1 & 2 \\ -1 & -2 \end{pmatrix}$

$A = \begin{pmatrix} -2 & 0 \\ 0 & -2 \end{pmatrix}$

$\begin{pmatrix} 4 & 5 \\ 0 & 0 \end{pmatrix}$

- (A) -1, -10 (B) 1, -10
(C) -1, 10 (D) 1, 10

$\begin{pmatrix} 1 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}$

If $z = 23y^2 + Sy$, then find

- (A) $4x+5y$ (B) $4x-5y$
(C) $4x+6y$ (D) $4x+6y +5xy$

6. $\begin{pmatrix} 2 & 2 \\ 2 & 2 \end{pmatrix}$

$\int \frac{1}{y} dy$

If y is an implicit function, then $\frac{dy}{dx}$

(A) $\frac{1}{y}$ (B) $\frac{1}{y^2}$

$\int \frac{1}{y} dy$

(C) $\frac{1}{y}$ (D) $\frac{1}{y^2}$

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7. If $f(x,y) = r'y \sin y$

$+ e$, then find $S(1,r)$

- (A) 2 (B) $27-e$
 (C) $27+e$ (D) 0

8. If u and v are functionally dependent, then their Jacobian value is 121

- (A) Zero (B) Positive
 (C) One (D) Negative

12 3 1

9. Solution of $(D^2 + 4)y = 0$

- (A) $y = (A \cos 2x + B \sin 2x)e^x$ (B) $y = A \cos 2x + B \sin 2x$
 (C) $y = A \cos 2x + B \sin 2x$ (D) $y = A e^{2x} + B e^{-2x}$

1131

10.

The value of

$D^{-1}a$

- (A) $x e^a$ (B) a
 (C) (D)

e

12 3 2

11. Solve $(D^2 + 5D + 4)y = 0$

- (A) $y = A e^{4x} + B e^{-x}$ (B) $y = A e^{4x} + B e^{-x}$
 (C) $y = A e^{4x} + B e^{-x}$ (D) $y = A e^{4x} + B e^{-x}$

12 3 2

12. Find the Particular Integral of $(D-4)y = \cos 2x$

- (A) $\cos 2x$ (B) $\frac{1}{8} \sin 2x$

8

- (C) $\frac{1}{8} \cos 2x$

$\cos x$

$\cos -$

$\sin 2x$ (D) $-\frac{1}{8} \sin 2x$

8 8

1 I4 1

13. The curvature of the straight line is

(B) 2

(A) 1

(C) 0 (D) -1

41

14. The envelope of the family of curves of the form $Aa + Ba + C = 0$, where

a is the parameter is

(A) $B + AC = 0$ (B) $B - 4AC = 0$

(D) $B - AC = 0$

(C) $B + 4AC = 0$

114 1

15. The radius of curvature in polar co-ordinates is

2

(A) $p = r + r^2 P^2 / -m + 2r$

(B) $p = -r^2 I^2 - * + 2$

(C) $p = -(r^2 T^2 / 2 - r^* + 2$

(D) $p = +r^2 / 2 + r^* + 2$

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3 21

that

.b. If $u = u(x, y)$ and $r = e^{\cos \theta}$ and $y = e^{\sin \theta}$ show

$\frac{\partial u}{\partial r} = \frac{\partial u}{\partial \theta}$

8

23. a. Solve $(D^2 + 2D + 1)y = e^x + \sin x$ in $2x$.

(OR) 32

D. Solve $(D + 1)y = \tan x$ by the method of variation of parameters.

3 4 2

Find the radius of curvature of the curve $r = a(1 + \cos \theta)$ at the point $\theta = \pi$.

(OR) 4 4 2

. Find the evolute of the parabola $y = 4ax$.

88 452

25. a.

Test the convergence of the series $\sum_{n=1}^{\infty} (V_{n+1} - V_n)^{-1}$.

(OR)

8 4 52

0. 2 2.4 2.4.62.4.6.8

Test the convergence of the Series $\sum_{n=1}^{\infty} \frac{1}{n!}$.

43.56 3.5.7.8 3.5.7.9.10

PART-C(1 x 15 = 15 Marks) Marks BL PO

Answer ANY ONE Question

26. Reduce the quadratic form $3x^2 - 2y^2 - z^2 - 4xy + 8xz + 12yz$ to canonical form by an orthogonal transformation. Discuss the nature of the quadratic form and also find rank, index, and signature.

form by an orthogonal transformation. Discuss the nature of the quadratic form and also find rank, index, and signature.

2

27. Find the dimensions of the rectangular box open at the top, of maximum capacity whose surface area is 432 sq.cm.

capacity whose surface area is 432 sq.cm.

*** *

===== SOURCE_FILE: nexus/Calculus And Linear Algebra/PYQs/pyq txt/PYQ Nov 2018.txt =====

CATEGORY: PYQs

STATUS: ok

27. Test the convergence of the series $\sum_{n=1}^{\infty} \frac{1}{n^2}$

"11@+1)Qn+1)

PART - C (5 x 2 = 60 Marks)

Answer AII B.Tech. DEGREE EXAMINATION,
Questions NOVEMBER 2018 .

First Semester

28. a) Solve the differential equation

using the method of variation of parameters.

Calculus-Hamilton AND LINEAR

; ; 11 ALGEBRA

theorem for the matrix A hence find A^{-1} . (For the candidates admitted during the academic year 2017-18)

^ t1 --iota1

(t1 2 t) Note: zeta

(D Part - A should be answered in OMR sheet within first 45 minutes and OMR sheet should be handed over to hall sheet

invigilator should be handed

(il) at the end of 45th minute.

Part - B and

b) Part - C should be answered in answer booklet.

Reduce the quadratic form $f(x, y, z) = x^2 + y^2 + z^2 - 2xy - 2yz + 2xz$ to canonical form and find the nature of the conic.

transformation. Time: Three

Also Hours

state its nature.- Max. Marks: 100

29' a Explain $e^{\cos x}$ in powers of x PART _Ae}xt=20Mayta)
 andy as far as the terms of the third degree. Answer ALL euestions

(tr4)

b' Find the volume of the larges, ,""[lfi* parallelepiped that l. The eigen values

*2 1" can be inserted in the of A2, given $e=1$ o z e I are
 ellipsoid $\{c*r\}-* b'' , ,!a.$

c, (A) 3,2,5 lo os)
 (c) 3,1,4 (B) 9,4,25

30' a' rnl g, r, re
 5o1r. 1o2 -1o+12)y=e5, +cos2x.

2. The characteristics
 oR) equation of the matrix ,=(1 , ,

b. solu" (*2D2+xD-g)y-+. - (
 (A) |

x' t2
 (c) -4a-s=0 (B) 5=s

A2 +4.x.+5=0 7z'a41_
 @) t", *qi_i__o

3 1 ' a' p;n6 the equation of the circle of curvature of the parabola $y^2 - 12x$ at the point
 (3, 6). (t 12\

3. The nature of the quadratic form, whose

b' matix is I t : f I i,
 rirra the evolute of the parabou ,z -(oR)

) 1, t6)
 (A

32. a-i. State the axioms of convergence of series by comparison test. pIndoefinsite(c)
 itivedennite El L::'ffJ#;*-"

ii. Test the convergence of the series $1 * 3.4 * 3.n.5 4'$

4 4.6 * ..-* - trth"canonicalfomtofaquadraticfor:nis
 4.6.8 - - '

(A)o -fi *fi+4rt,thenthesignatureof

(c) $2(B)^3 e^{Fis}$

iDi 1

b.i. State the Cauchy's criterion

"r"orro!:il" of an infinite series.

ii. 5. If $u = y^2 + 2y$, where $x : a^*$, $y : Z_{at}'tdhtat4$
Test the conyergence of the series

,Ir(fffi W) Ii] z,?t21t+st1 di
(C) $r, r, r(r*r,)$

4a3t378,+5,ty (D) 0oz'riltiSr>
**:f,S*

6' Th" stationary points of the function
 $2,0) 3*2 - y^2* 13 *$

(A) $(0, 0), (- (B)$
(c) $(0, 0), (2, 0)$

$(1, 1), (_2,0)$
Page 4 of4 (D) ir, ri.d,-o)

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16. When n is 6 *ve integer, $F(n + 1) =$
(A) $(n+1)!$ (B) $n!$

(D) $(n-1)!$
7. The x spect to'' e (c)

$= r \cos; , y = r \sin Le$,then the Jacobian of r, y with re (;'" (*)) $Qn)'!$

(A) r (B) $lr \ col$
(C) $2r @\ 2lr$ 17. The series $F -lf$ is convergent if

u
 $n - \ln'$

8. Maclaurin (A) $p >$ (B) $p < 1$
1

series for error is

$t^* = \frac{1}{3} (x - \sqrt{3} (c) p: I @) p: o$

(A) $1 + (x+y) - (x^2) + \dots$ (B) $1 - (x-y) - \dots$

3t 111

. $(x + \sqrt{3})^3 + 18$. The series $1 + \dots$ is

(C) $1 + (x-y) - (x^2) + \dots$ (D) $1 - (x-y) - \dots$

(A) Convergent (B) Oscillating

(C) Divergent (D) Monotonic

9. The solution of $(x^2 + 4)y' = 0$ is

(A) $y = \cos x + \sin 2x$ (B) $y = \cos x + \sin 2x$

$y = A \cos x + B \sin x$ 19. As per D'Alembert's ratio test, if $\lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = L$ is a series of positive terms and $L < 1$

(c) (0) $y = \cos x + \sin x$ if $t = 0$

$y = A \cos x + B \sin x$

series is convergent, if

10. The particular integral of the differential equation $(x^2 - 10x + 25)y'' + \dots = 1$ is (A) $\frac{1}{x}$ (B) $\frac{1}{x^2}$

1

(A) (B) (c) $\frac{1}{x}$ (D) $\frac{1}{x^2}$

$(\cos x + 3 \sin x) \ln(\cos x - 3 \sin x)$

1

$(\cos x + 3 \sin x) \ln(\cos x - 3 \sin x)$

$5 \cos^3 x - 3 \sin^2 x$ (D) $\cos x - 3 \sin x$

(C) Divergent (D) Absolutely convergent

11. The complementary function of $(D^2 + 4D + 4)y = 0$ is

e^{-2x}

(A) $cr = (Ax + 8.1e^{-2x})$ (B) $gp = 7Ax + B1e^{-2x}$ PART-B (5x 4=2A Marks)

) ^

(C) $CF = \cos x + \sin 2x$ (D) $CF = 7e^{-2x} + Be^{-4x}$ Answer AI{Y FM Questions

$(-t)^3$

12. The complementary function of $(x^2 + ax + 2)t = 2x^2 + 21$. Use Cayley-Hamilton theorem to find a , given that $n = 1, 4$

(A) $y = 7Ax + B)e^{-zx}$

(C) $y = Ae^{-x} + Be^{4x}$ 22. $\frac{d^2y}{dx^2} + \frac{dy}{dx} + y = 0$, at $x = 0, y = 1, \frac{dy}{dx} = 0$

" $\frac{d^2y}{dx^2} + \frac{dy}{dx} + y = 0$

13. The radius of curvature of the curve: $y = x^2 + 2x + 1$. Find x_1, x_2, x_3 if the Jacobian of x_1, x_2, x_3 with respect to

(A) $2x$,

(C) $2x^2 + 4x + 1$ 24. Find the eigenvalues of the matrix $A = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$

14. The radius of curvature of a circle at any point is same as its

(A) chord $(2 - z)$

(C) Diameter $2r$ 24. Find the eigenvalues of the matrix $A = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$

$u_3 = 1$

15. The value of $\sin^{-1}(\frac{1}{2})$ is

(A) $\frac{\pi}{6}$ 25. Solve the equation $10x^2 - 10x + 1 = \sin^3 x$

4

(C) n 26. $y = \sin x$

Find the envelope of the family of straight lines given by $x \cos \alpha = a \sec \alpha$

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